

The Grammar of Graphics

DATA 3302: Data Visualization

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Cal Poly San Luis Obispo · Fall 2026

Slides based on:

Munzner, Visualization Analysis and Design, 2014

Bertin, Semiology of Graphics, 1967

Cleveland & McGill, 1984

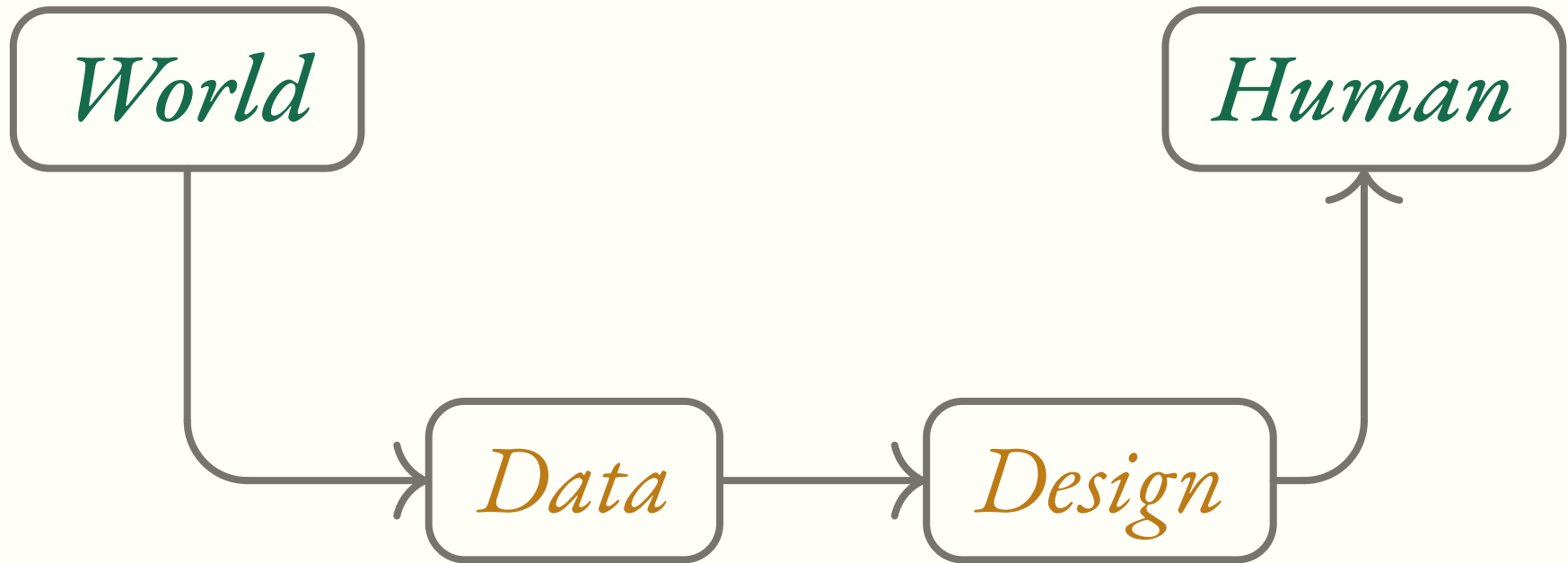
Polo Chau, Data and Visual Analytics Course Slides.



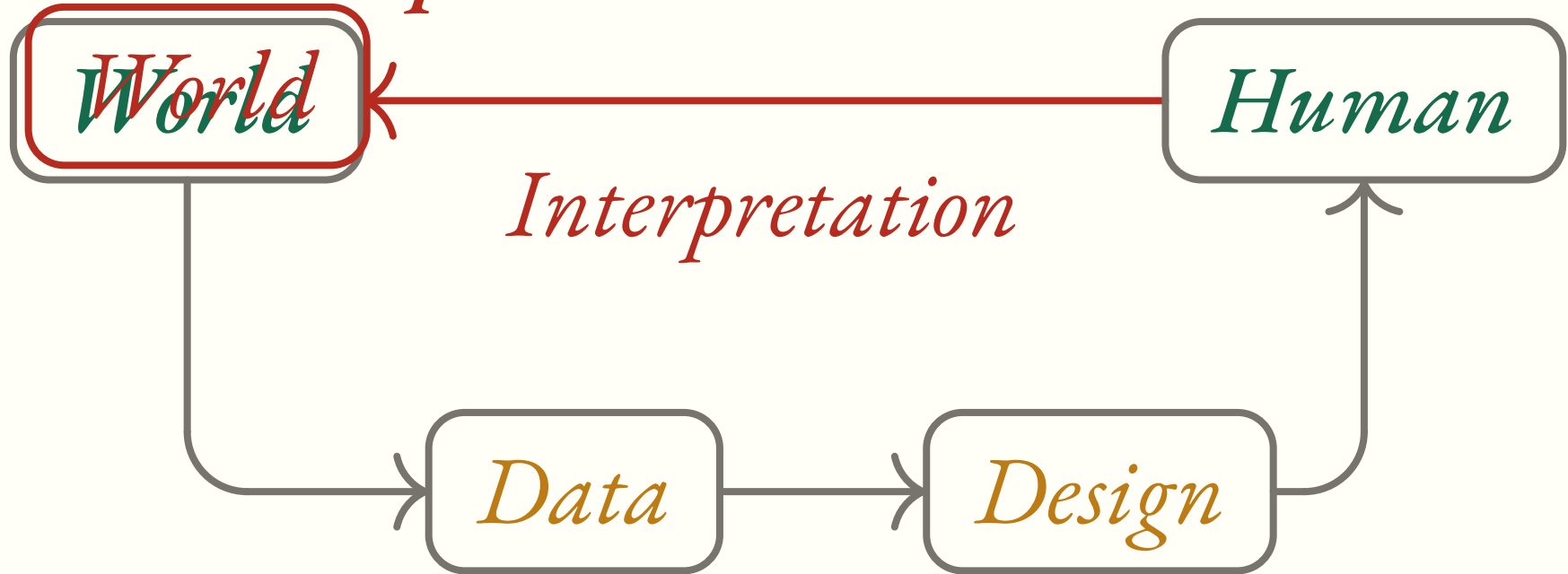
What is a visualization?

Recall

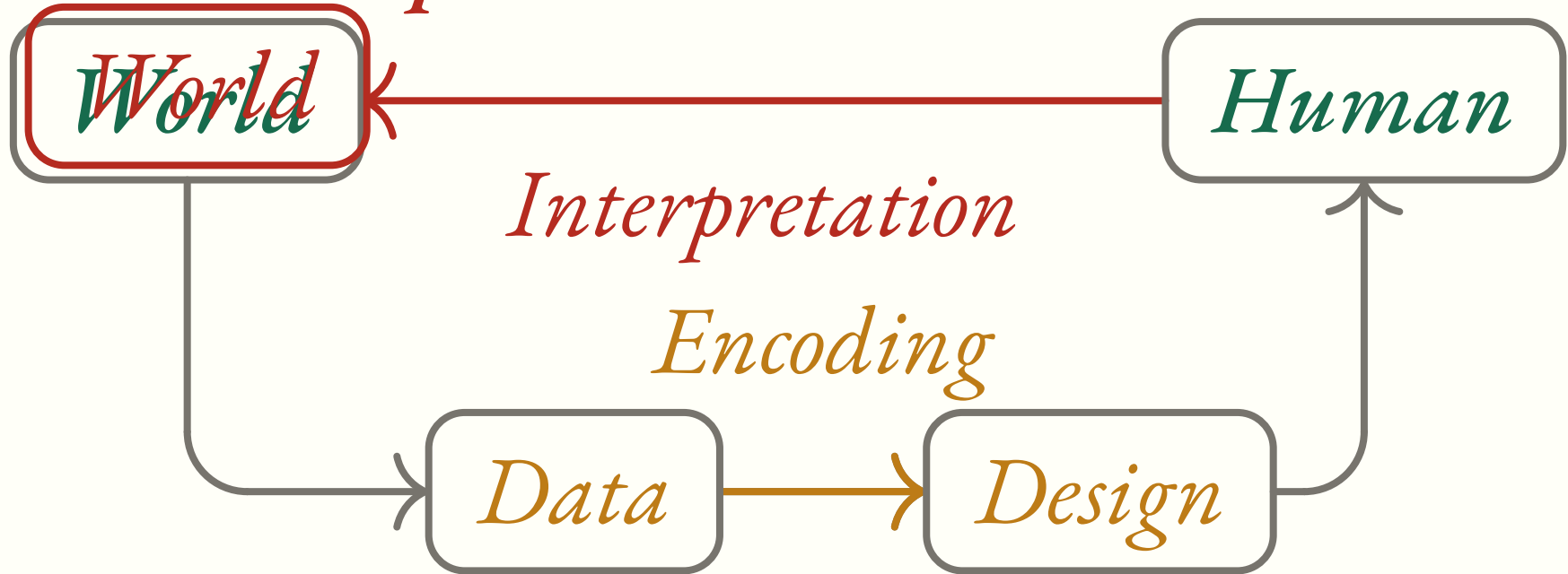
*The study of mapping **data elements** to **visual elements** to clearly communicate **true** and **interesting** properties of data.*



Mind the Gap!



Mind the Gap!



*A **visual encoding** is a mapping
from data to visuals.*

Data

Quantitative : value / amount

Categorical : kind of thing

Processing : transform, aggregate

Data

Quantitative : value / amount

Categorical : kind of thing

Processing : transform, aggregate

— *Encoding* ➔

Visual

Graphical marks : points, lines, areas

Channels : position, size, color ...

Glyphs : composite marks

The Grammar of Graphics

A graphic is built by binding data attributes to the visual properties of geometric marks.

A Grammar for Graphics

Instead of memorizing named chart types, compose a few primitives:

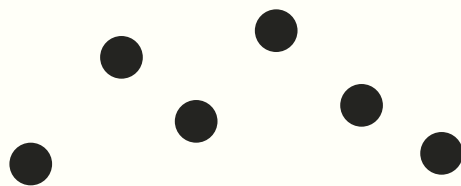
1. Pick a *mark*, some visual item
2. Bind each data attribute to a *channel*, a visual property of the mark
3. *Compose* marks into layers, facets, and coordinate systems

Wilkinson's Grammar of Graphics (2005) forms the structure behind libraries such as ggplot2, and Vega-Lite.

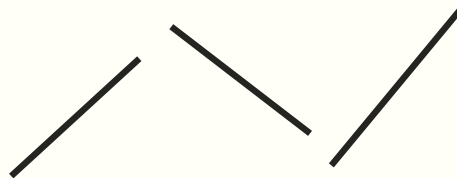
Marks

The geometric primitives of a graphic.

Marks: Geometric Primitives



Points

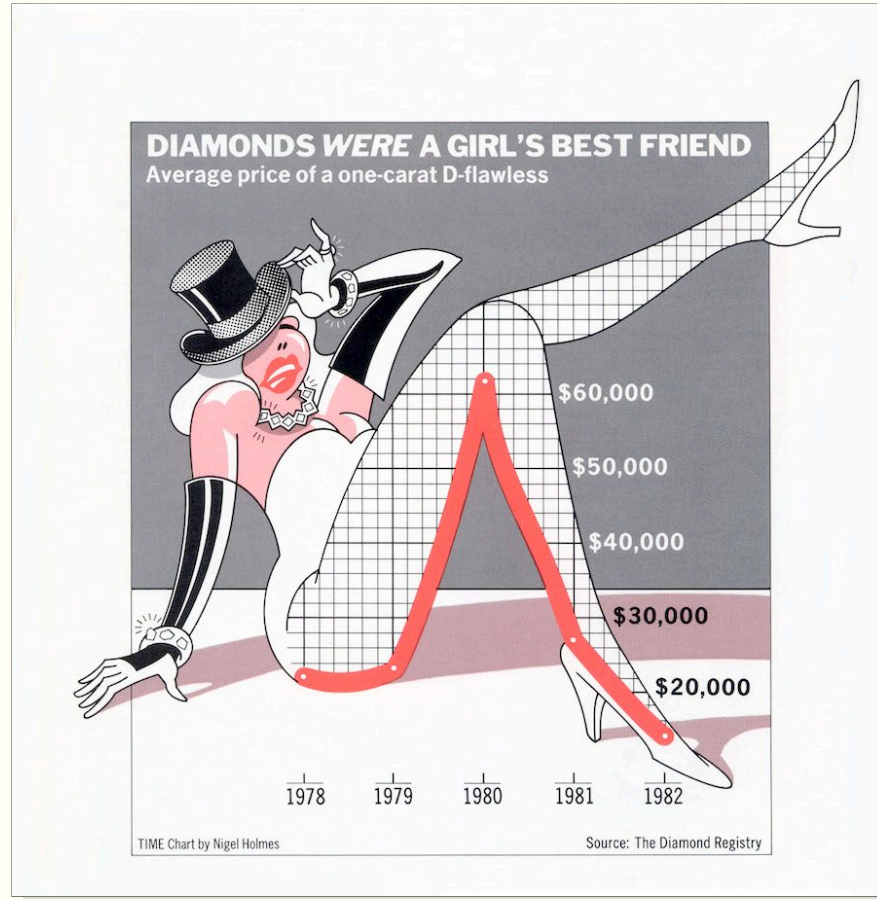


Lines



Areas

Marks have dimensionality: points (0D), lines (1D), areas (2D), and volumes (3D).



Chartjunk: Extra stuff that does not correspond to an encoding, lowering the data-to-ink ratio

Channels

The visual properties that can vary for a mark

Channels

The visual properties that can vary for a mark

Mapped to by properties of corresponding data

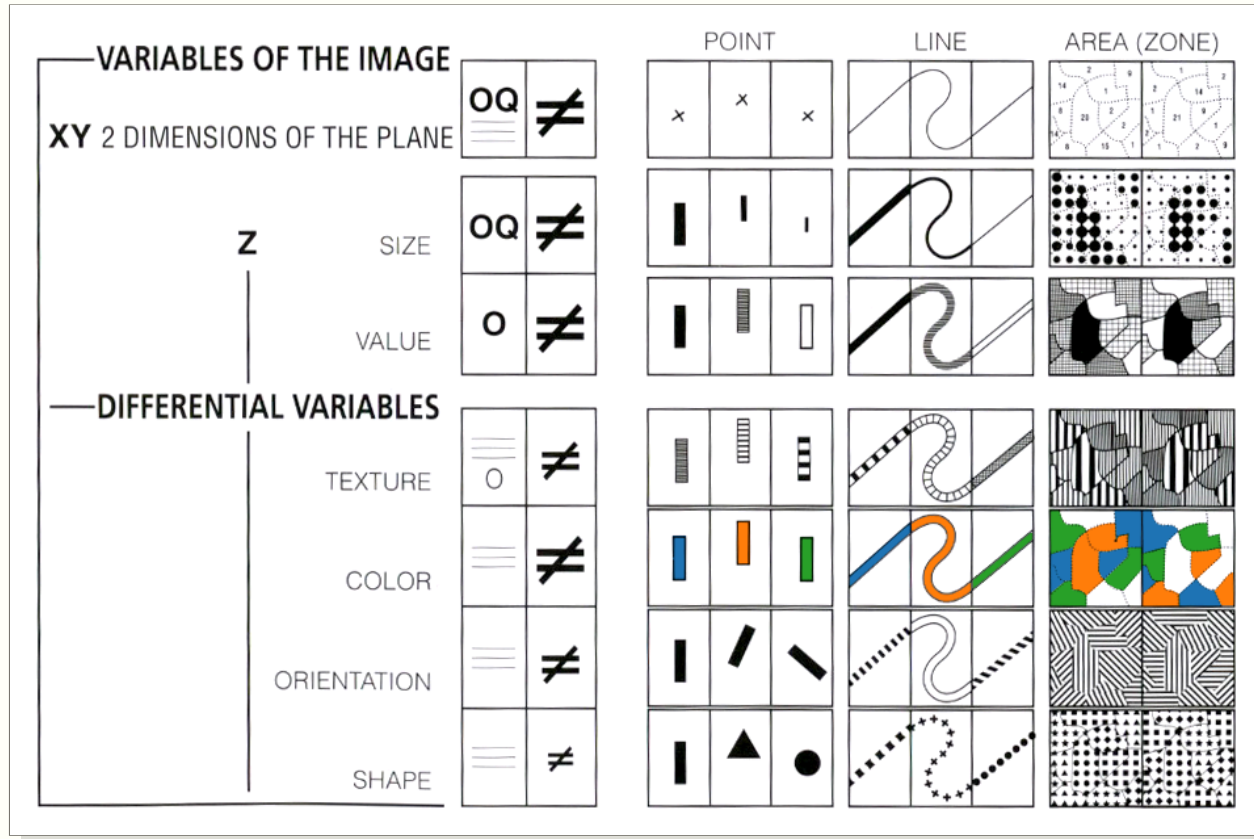
Primitive Channels

1. Position
2. Size
3. Value / luminance
4. Texture
5. Color
6. Orientation / tilt
7. Shape

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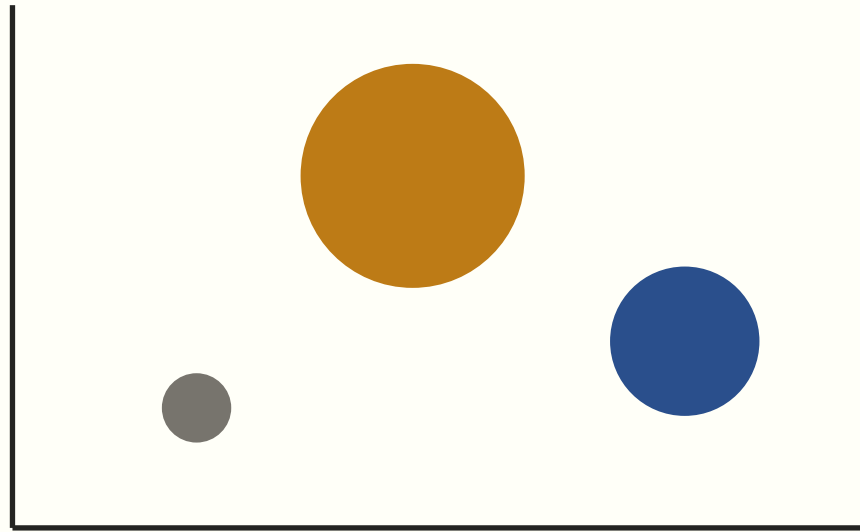
Bertin classified each channel by what the eye can read from it: *order*, *grouping*, or *exact quantity*.

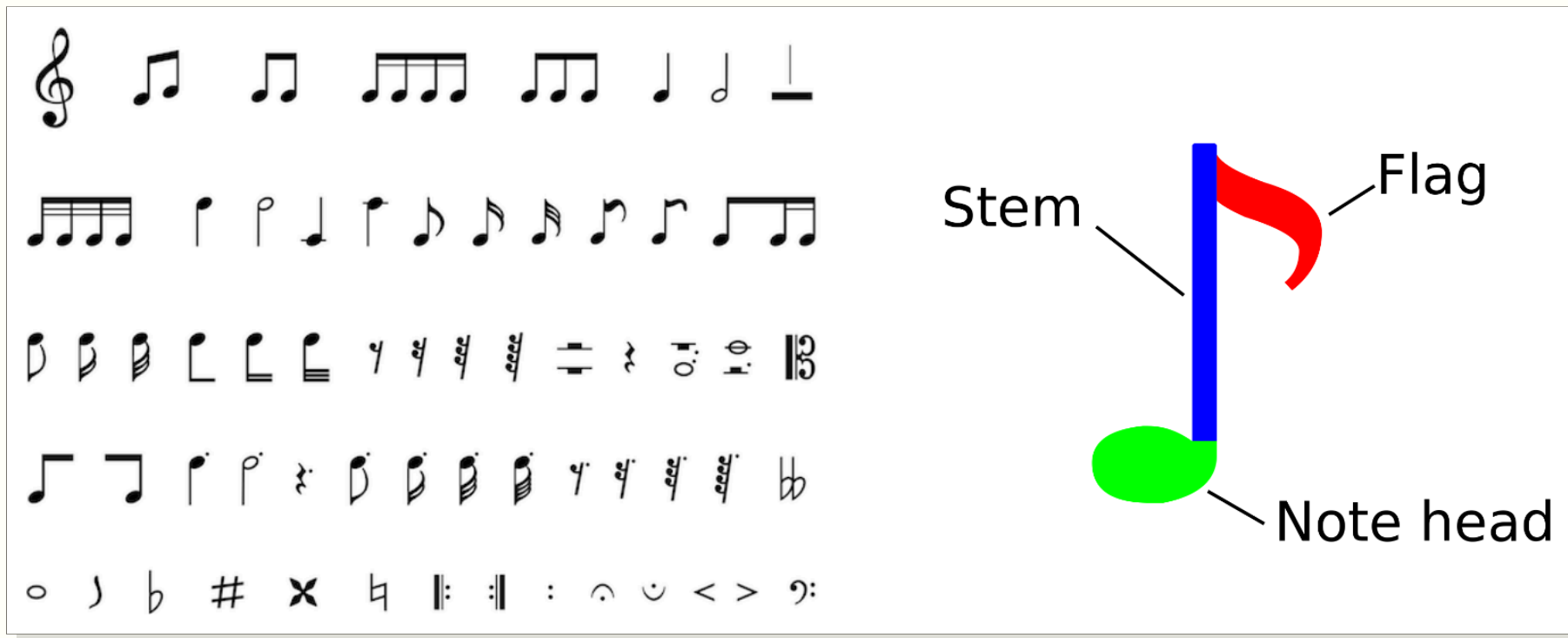


The “retinal variables,” each varied across point, line, and area marks.

Jacques Bertin, *Semiology of Graphics*, 1967.

How many / which channels are being used?





Symbolic channels: learned systems where stem, flag, and note-head each carry meaning.

Standard music notation.

Channels express one of two things:

How much? (magnitude or order)

Which one? (identity or category)

➔ Magnitude Channels: Ordered Attributes	➔ Identity Channels: Categorical Attributes
Position on common scale	Spatial region
Position on unaligned scale	Color hue
Length (1D size)	Motion
Tilt/angle	Shape
Area (2D size)	
Depth (3D position)	
Color luminance	
Color saturation	
Curvature	
Volume (3D size)	

Magnitude channels carry ordered attributes; identity channels carry categorical ones.

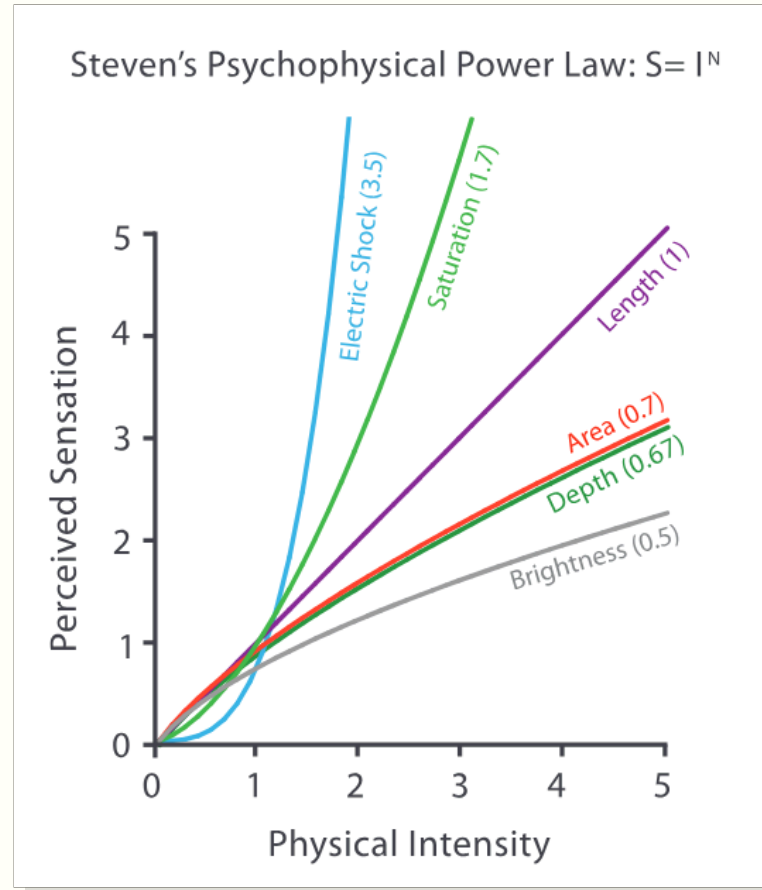
Munzner, Visualization Analysis and Design, 2014, Fig. 5.1.

Two tests for a channel:

*Is it **expressive**?*

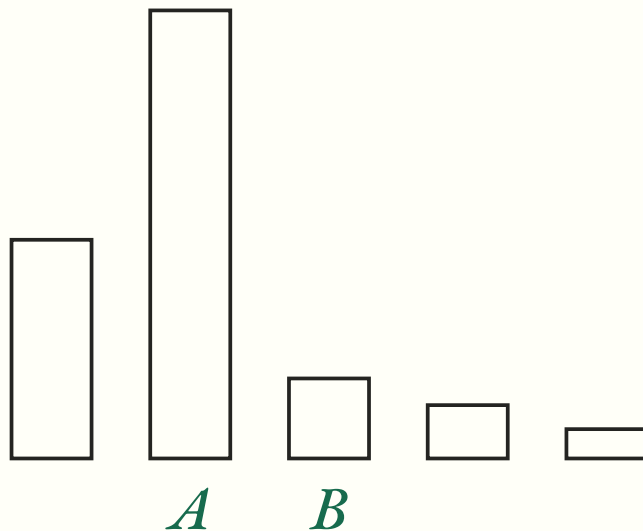
*Is it **effective**?*

*Don't just mind the gap ... **measure** it!*

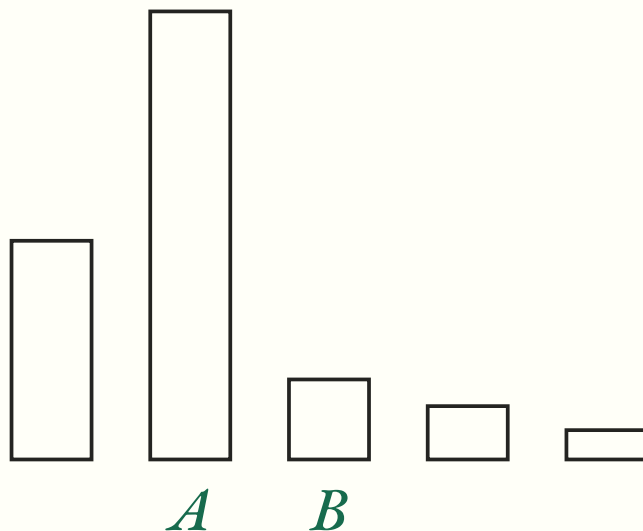


Perception is different than objective encoding

Length: how much taller is bar *A* than bar *B*?

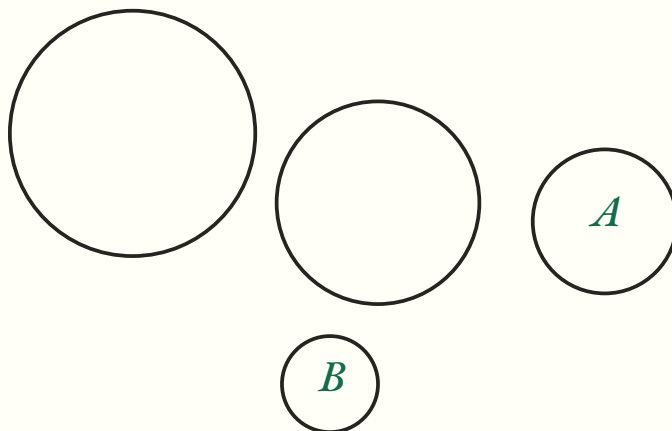


Length: how much taller is bar *A* than bar *B*?

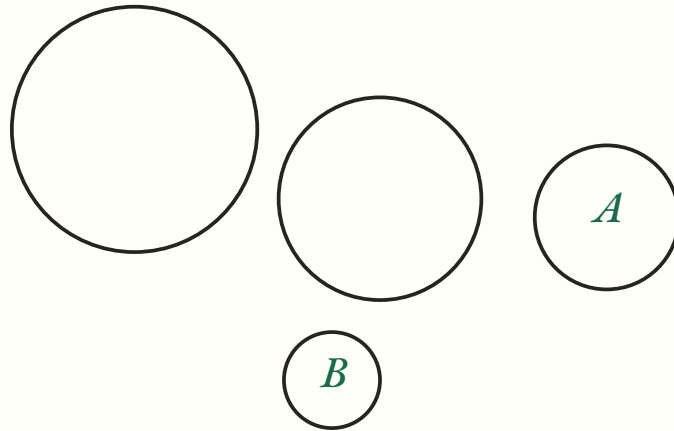


A is about 5.6× taller.

Area: how much bigger is circle *A* than circle *B*?

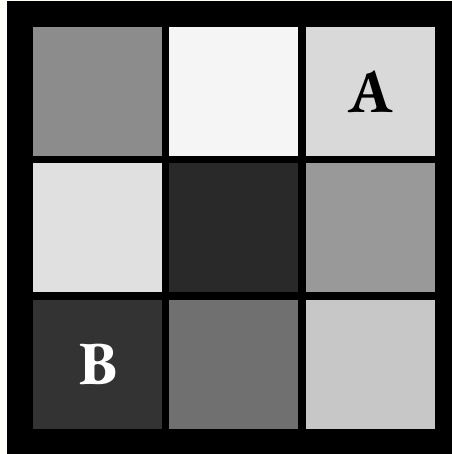


Area: how much bigger is circle *A* than circle *B*?

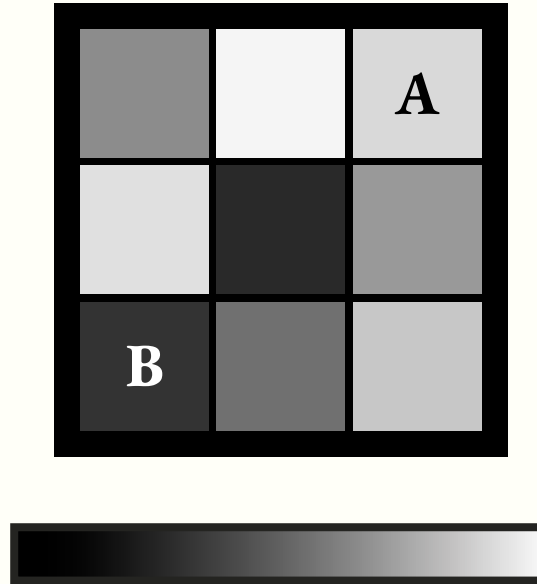


A has about 2.25× the area

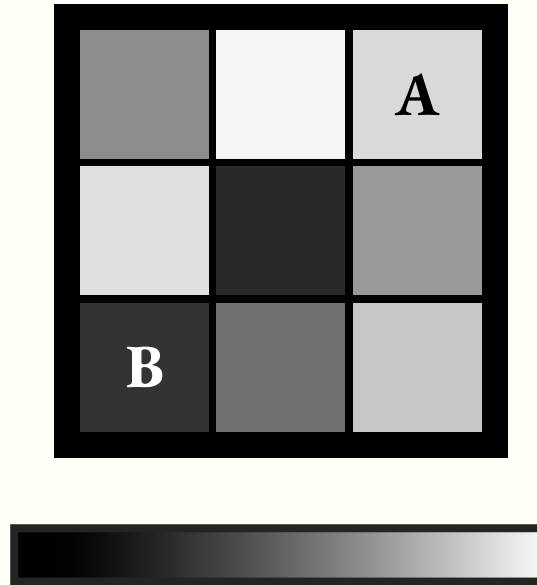
How much brighter is square *A* than square *B*?



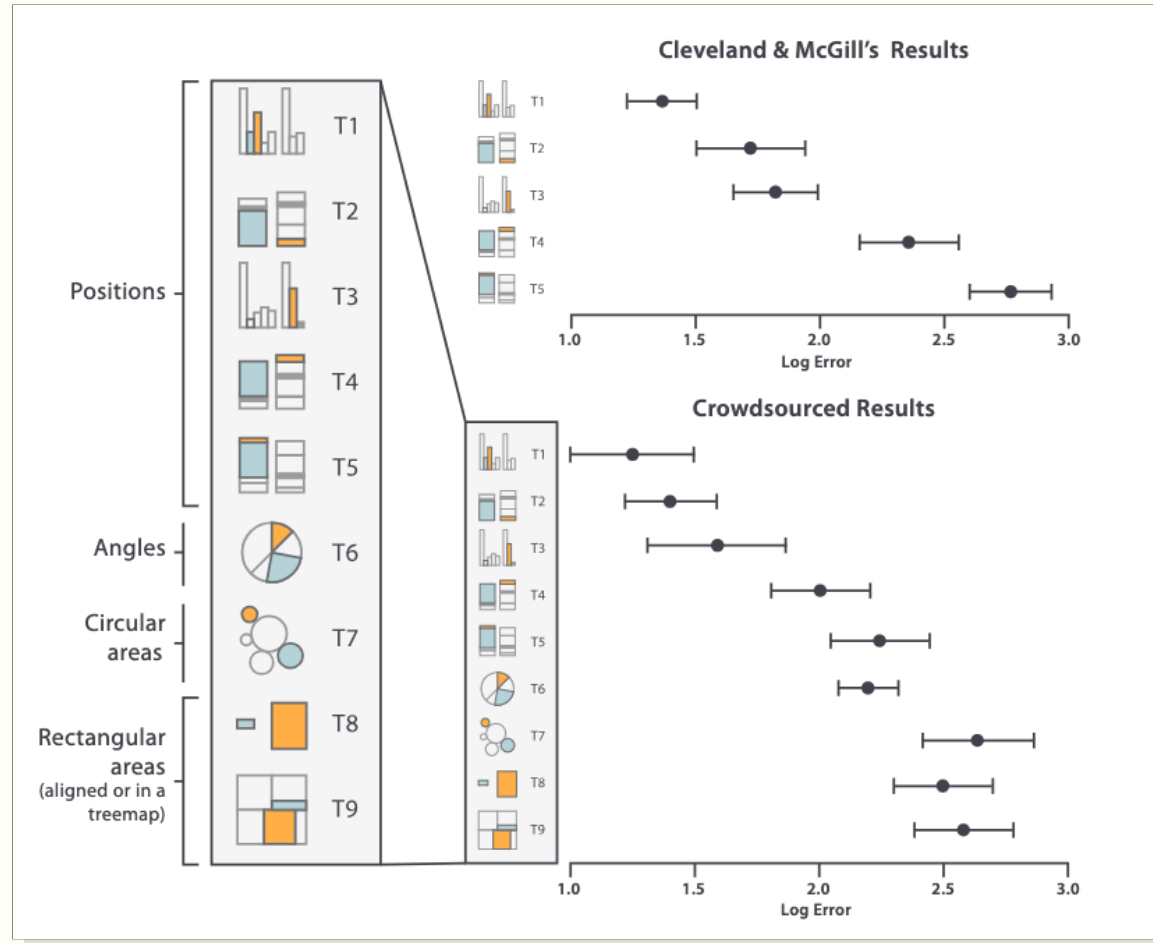
How much brighter is square *A* than square *B*?



How much brighter is square *A* than square *B*?



A is about 4.25× the luminance of B.



Cleveland & McGill, 1984; Heer & Bostock, 2010.

Position channels work when...

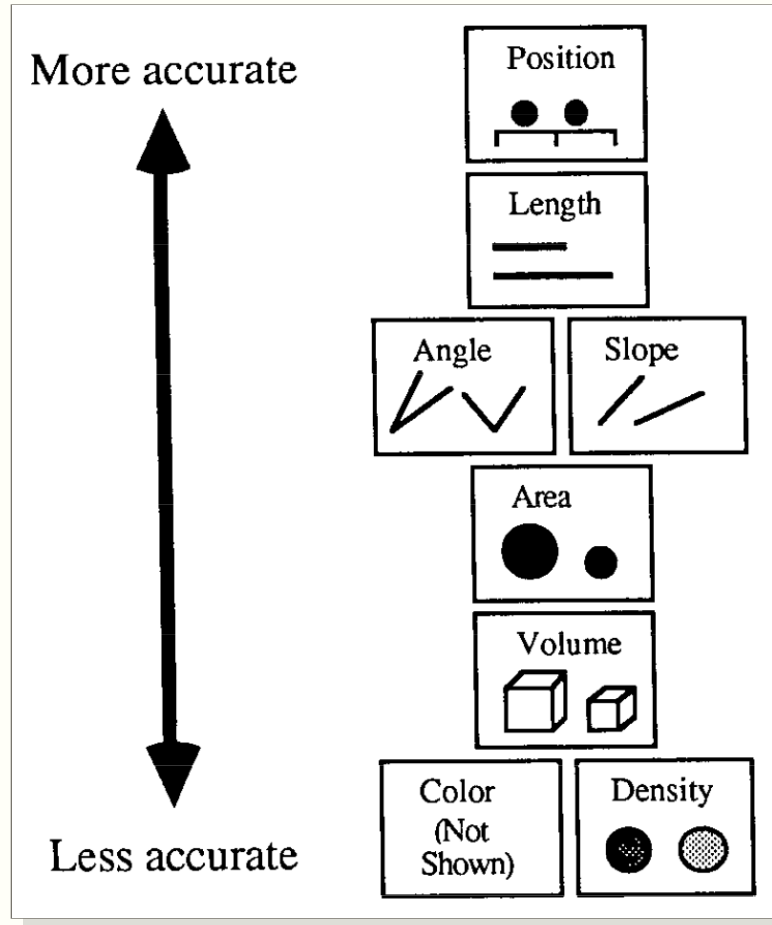
1. One spatial dimension
2. Aligned to a common scale
3. Marks kept in proximity
4. Linear scale

Position channels work when...

1. One spatial dimension
2. Aligned to a common scale
3. Marks kept in proximity
4. Linear scale

...and fail when

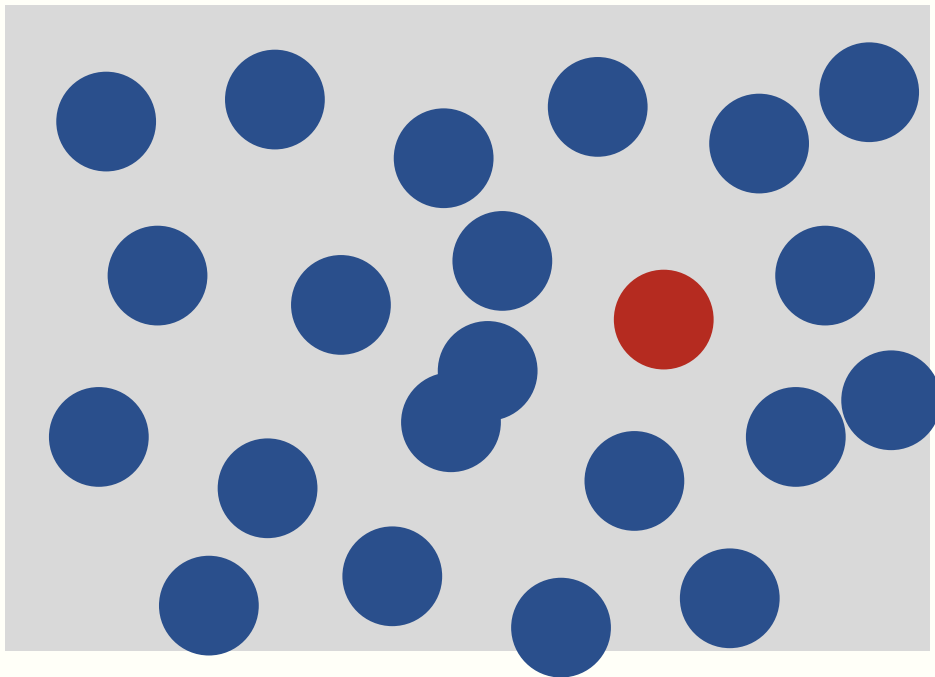
1. Two plus dimensions at once
2. Baselines misaligned
3. Marks scattered apart
4. Scale non-linear



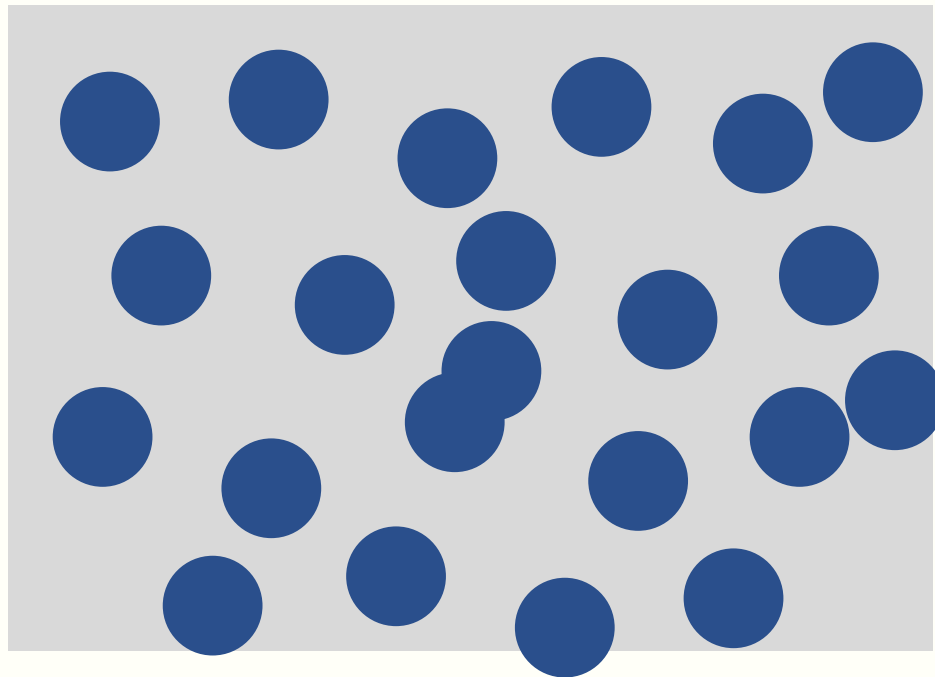
Channels ranked by judgment accuracy for quantitative data.

Channels are not read in isolation

*Raise your hand when you see the **red dot**.*



Left

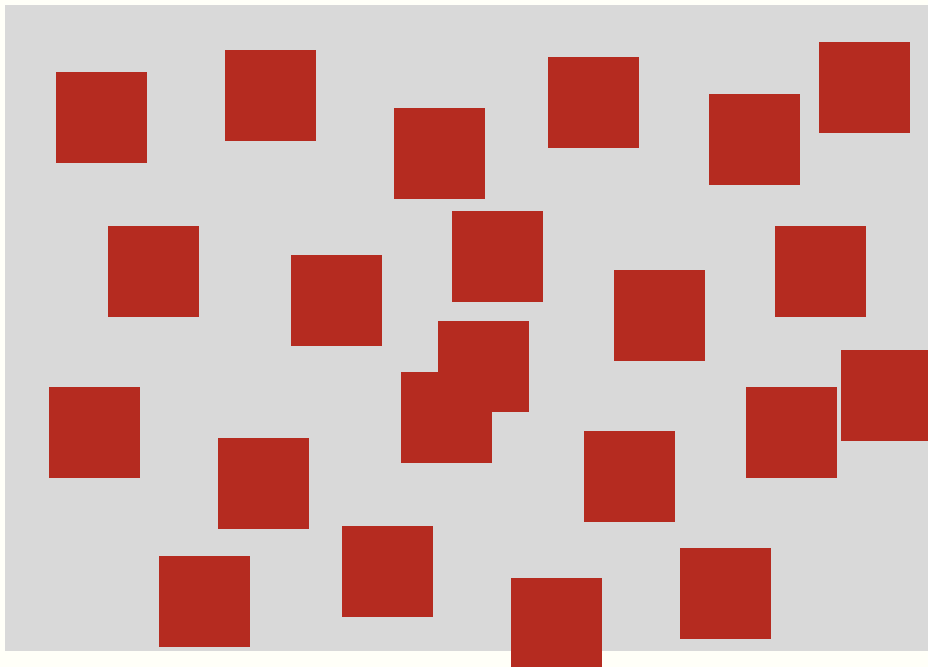


Right

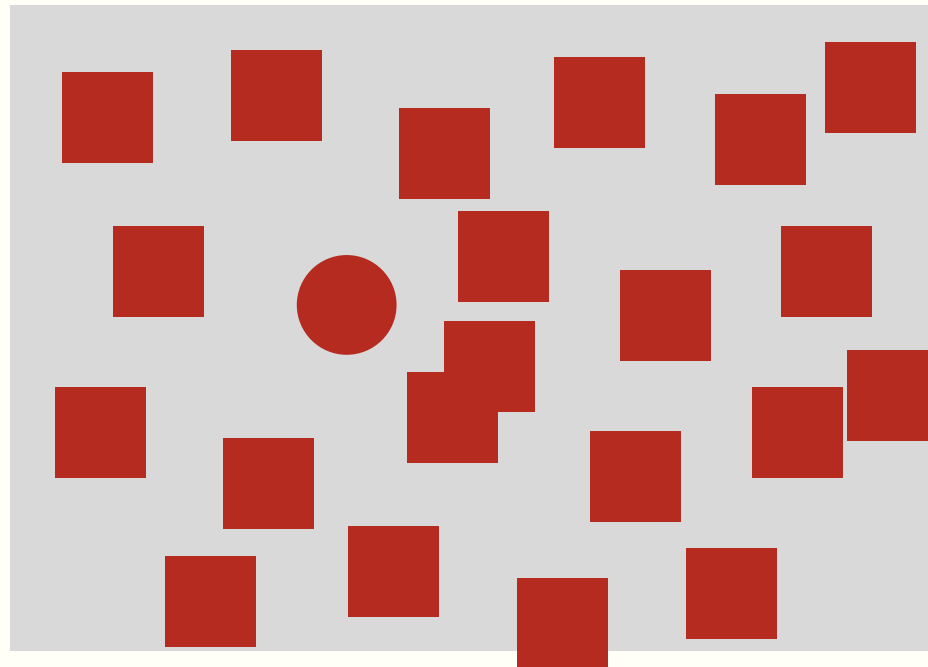
Which side was it on?

Was it easy/fast to tell?

*Raise your hand when you see the **red dot**.*



Left

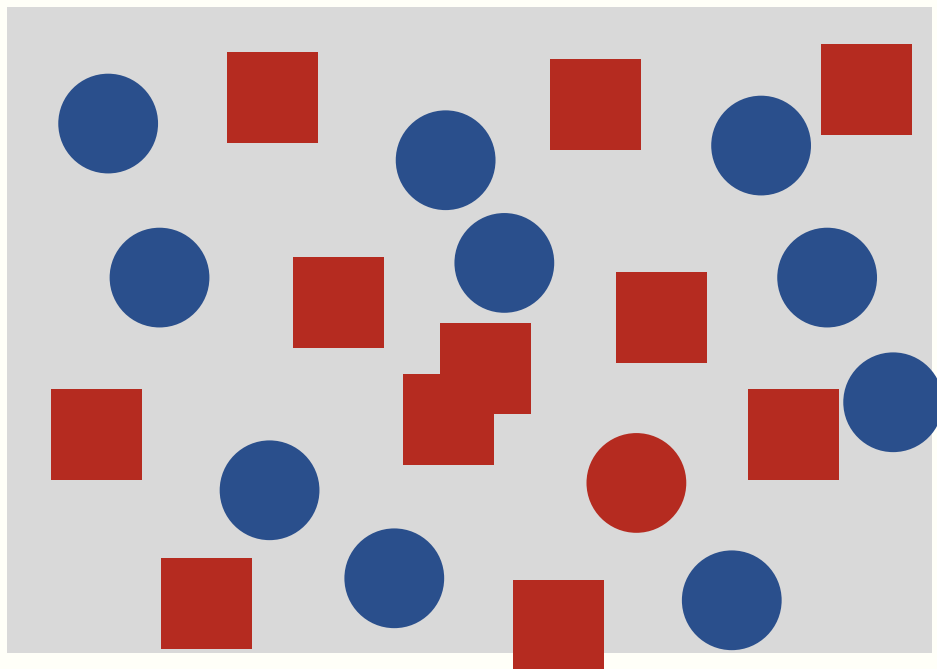


Right

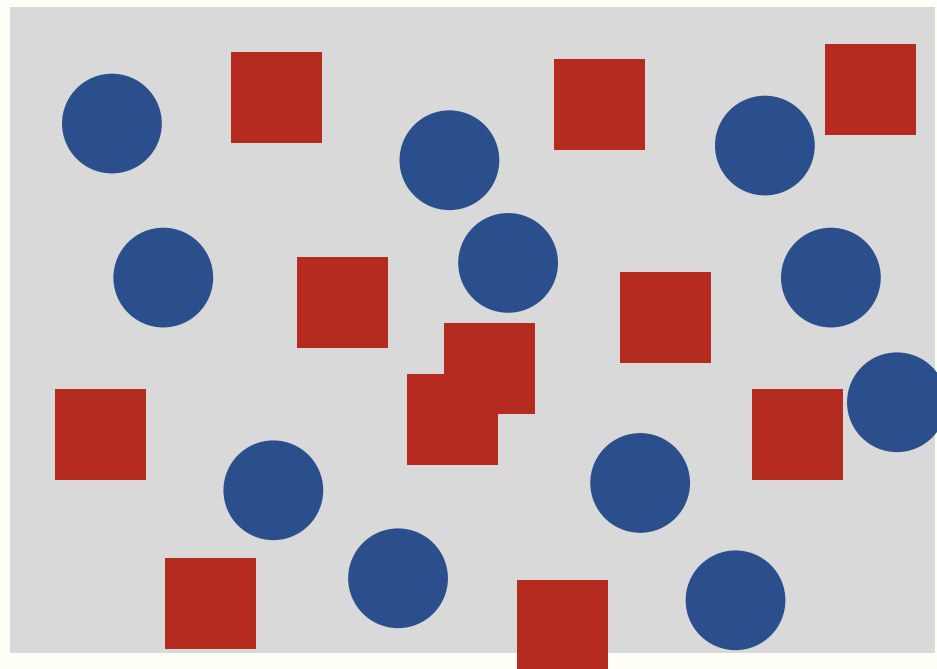
Which side was it on?

Was it easy/fast to tell?

*Raise your hand when you see the **red dot**.*



Left



Right

Which side was it on?

Was it easy/fast to tell?

A conjunction of two channels forces a serial search.

An encoding is marks + channels.

A good design utilized marks and channels the eye can actually read.

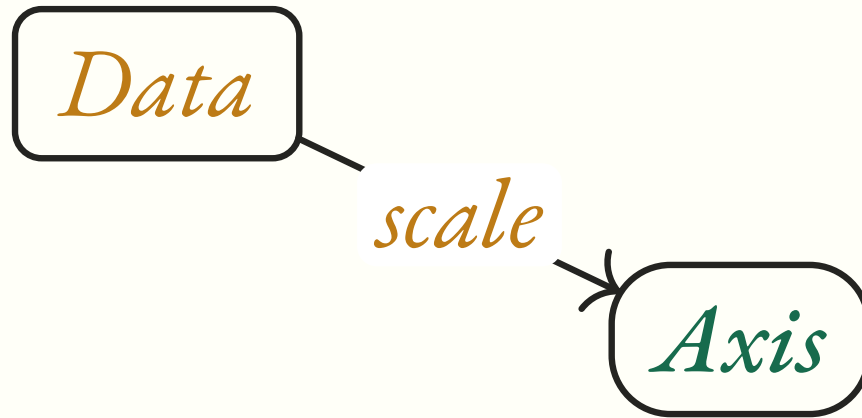
*How do we actually **convert** data attributes to visual channels?*

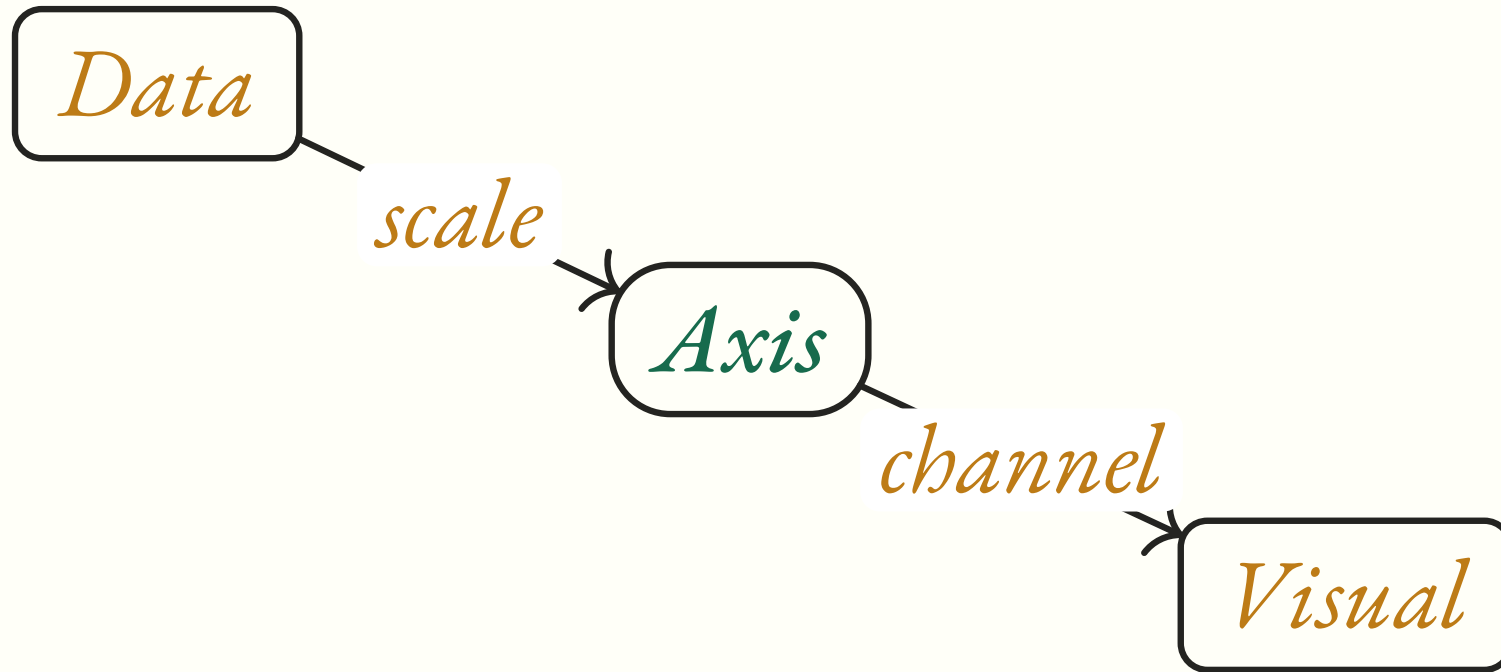
Scales

Scale

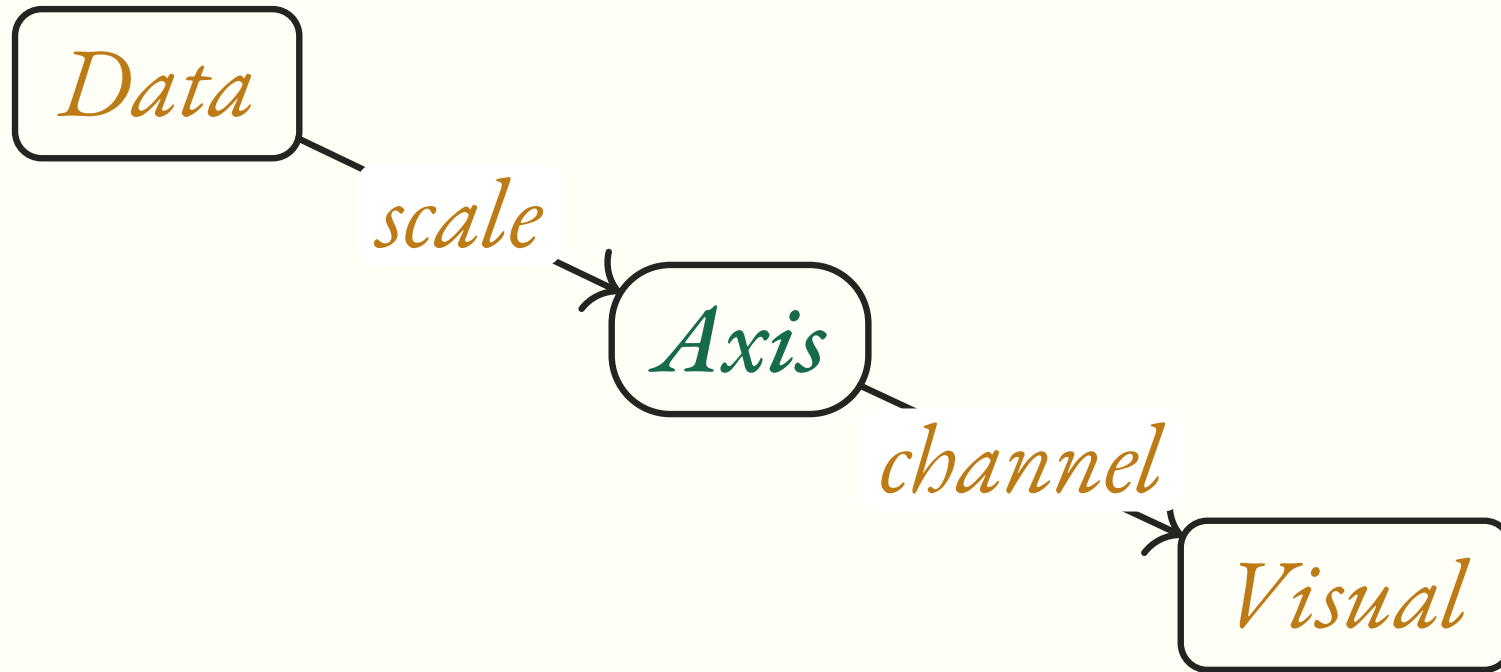
A *scale* is a mapping from a *domain* (values in the data) to a *range* (values in the display).

Every channel you bind in an encoding gets a scale.





Encoding



Scales Are Functions

Domain → *Range*

abstract dimension → *channel input*

size of army in people → width of bar in px

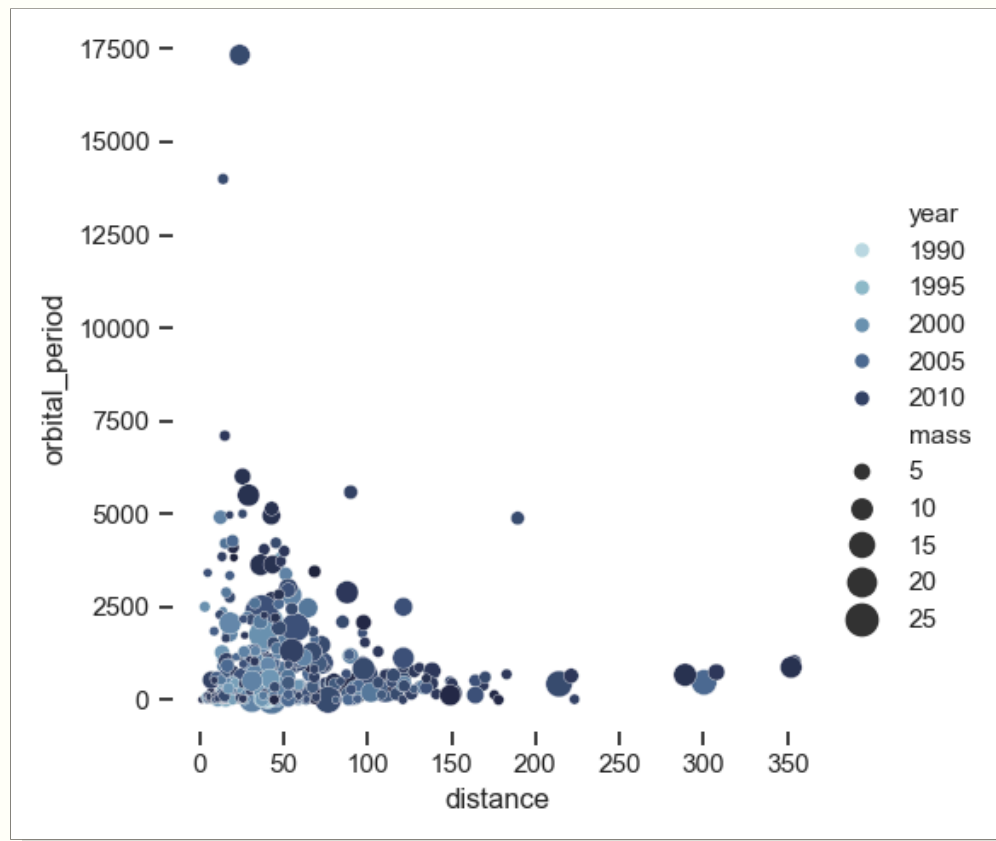
What properties are important for scale functions?

Linear scales: $v_{\text{axis}} = v_{\text{data}} \cdot \text{scale} + \text{bias}$

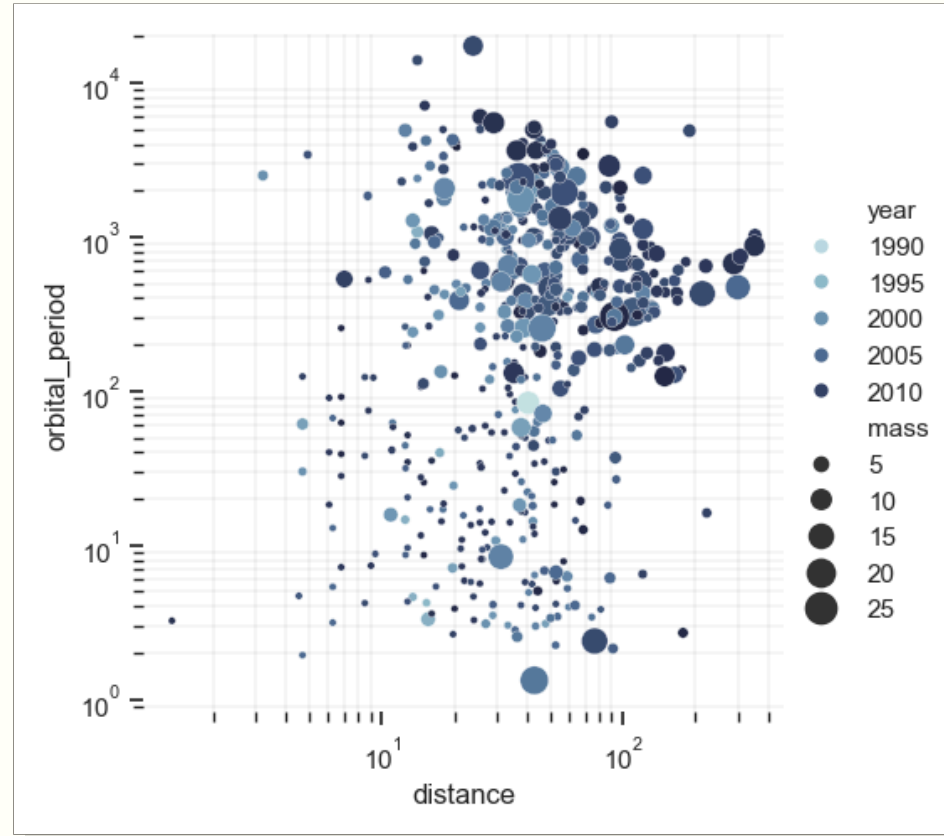
- *Equal data steps map to equal visual steps*
- *Preserve proportionality*
- *Sensible default*
- *Issues in certain distributions*

Log scales: $v_{\text{axis}} = \log_{\text{base}}(v_{\text{data}}) + \text{bias}$

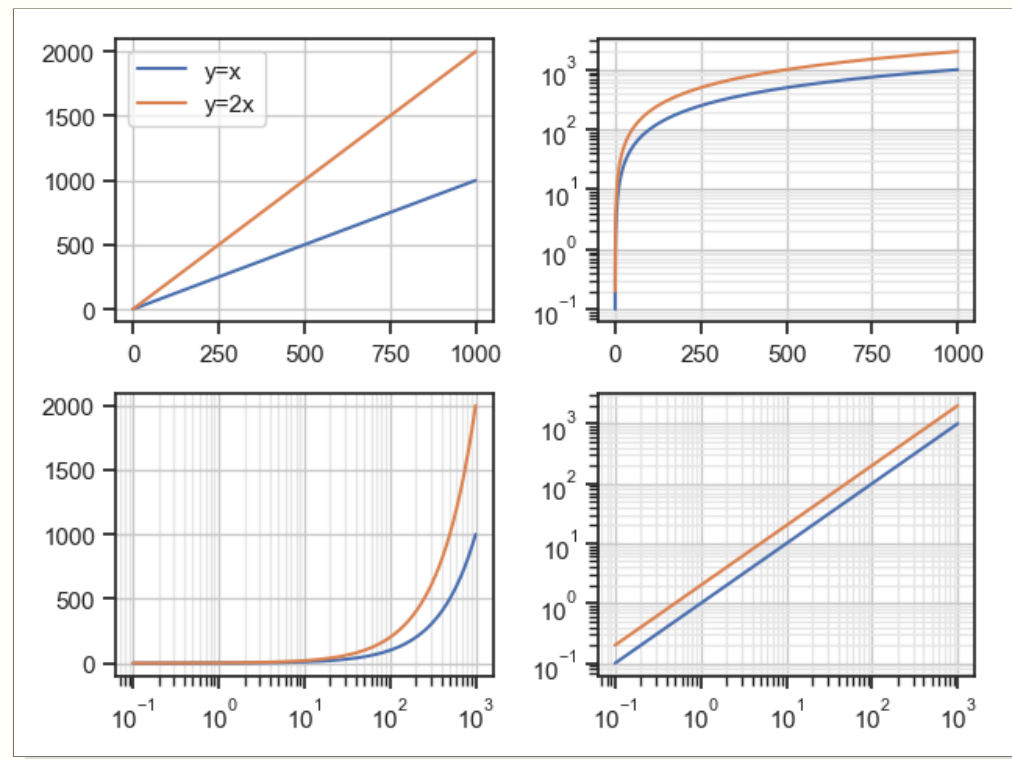
- *Equal visual steps map to multiplication by base*
- *Long tail distributions*
- *Mixing very big and/or very small numbers*



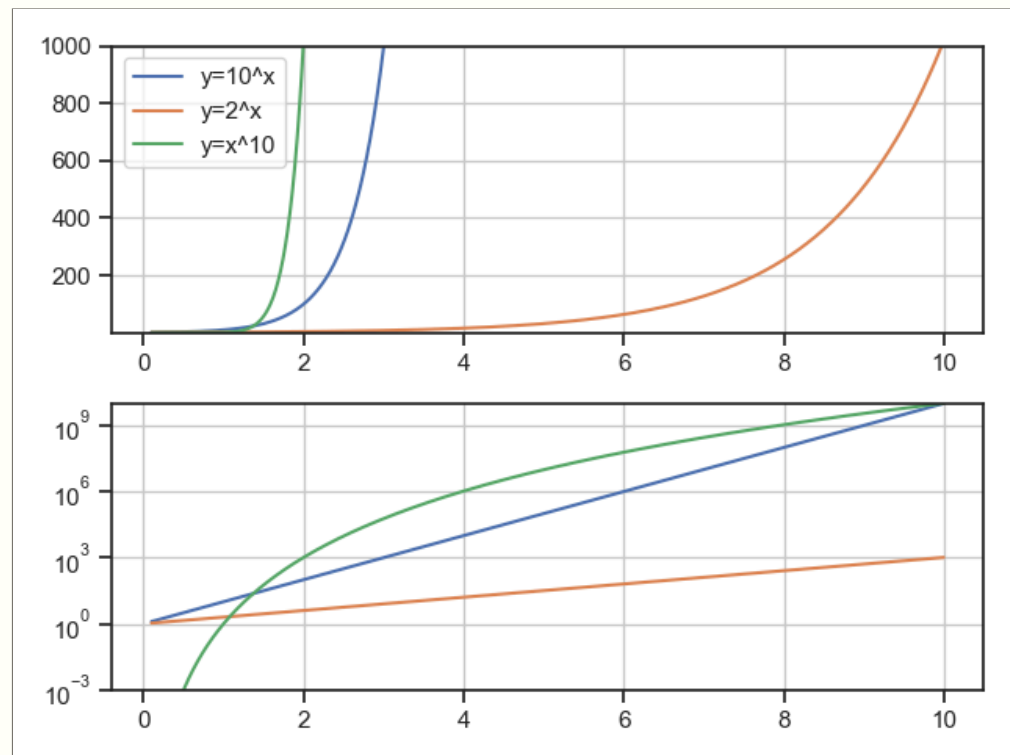
*Exoplanets on **linear** scale: everything piles into one corner.*



*The same data on **log** scale: note the grouped minor ticks.*



$y = x$ and $y = 2x$ under linear and log axes: a log scale hides a constant factor.



On a log y-axis exponentials become straight lines and their base becomes a slope.

$$v_{\text{axis}} = \log_{\text{base}}(v_{\text{data}}) + \text{bias}$$

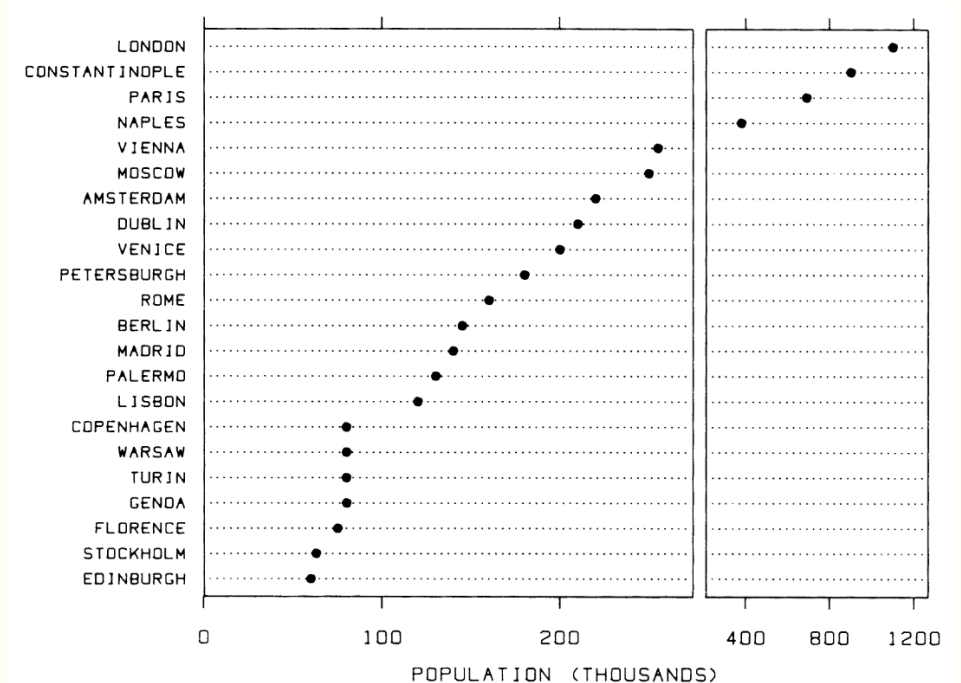
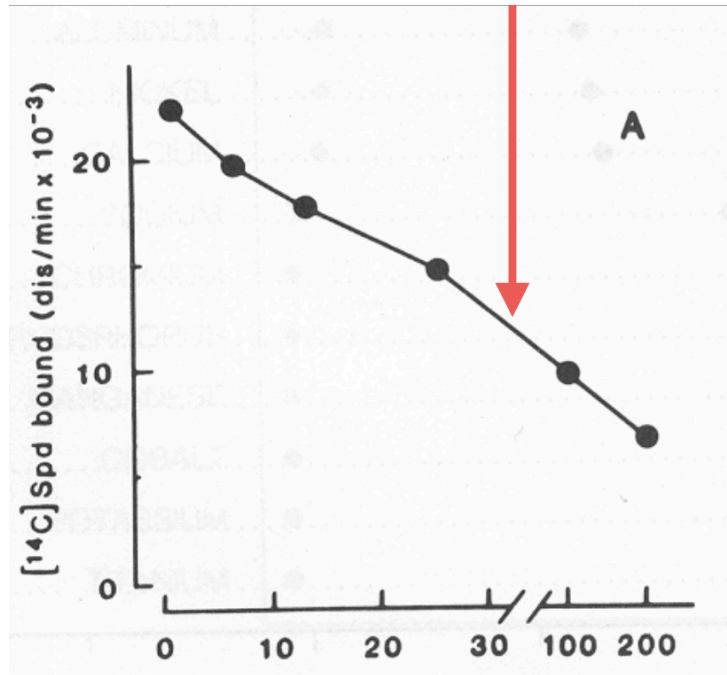
What about $v_{\text{data}} = 0$?

$$v_{\text{axis}} = \log_{\text{base}}(v_{\text{data}}) + \text{bias}$$

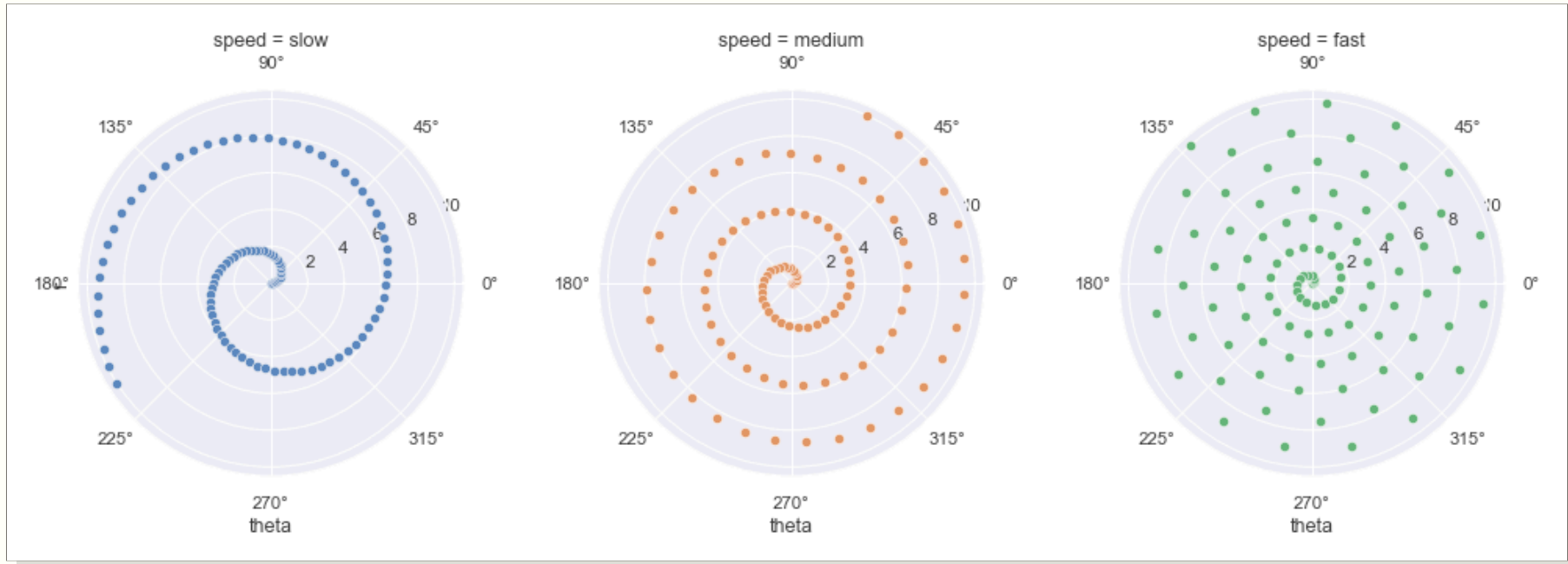
What about $v_{\text{data}} = 0$?

$$v_{\text{axis}} = \log_{\text{base}}(v_{\text{data}} + \varepsilon) + \text{bias}$$

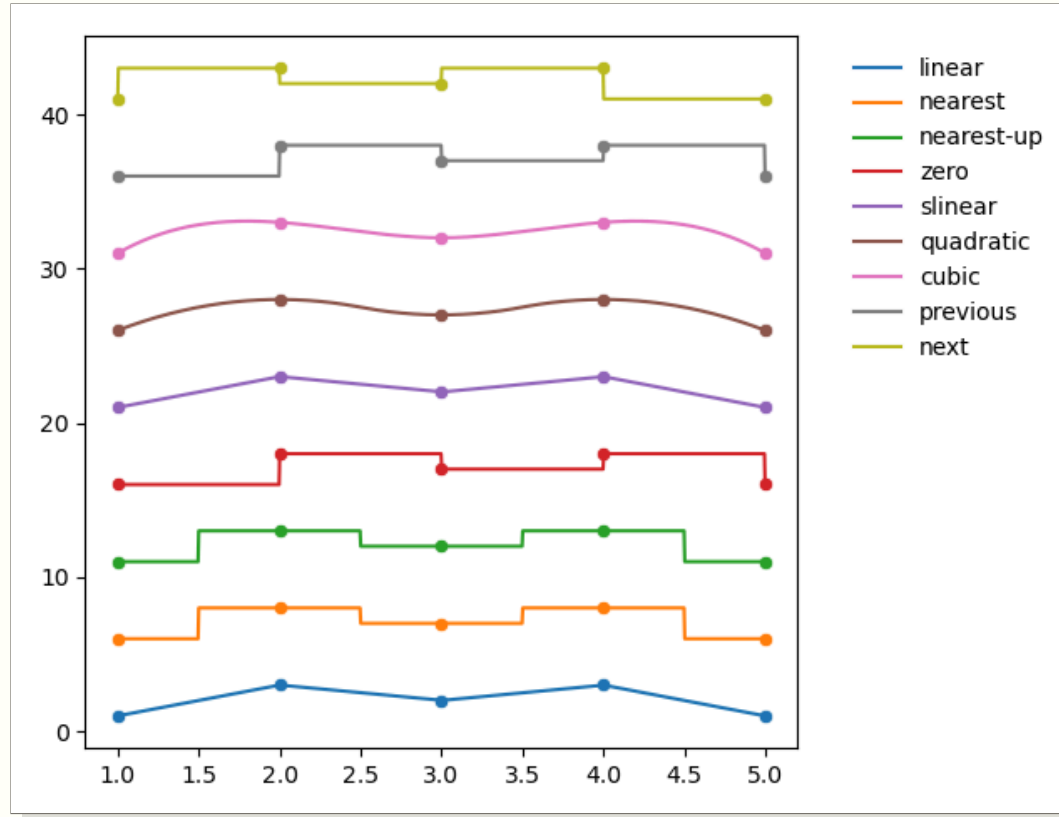
Scale Breaks: Handle With Care



A break in the axis breaks the encoding. Prefer a log scale, or split into panels if you must.

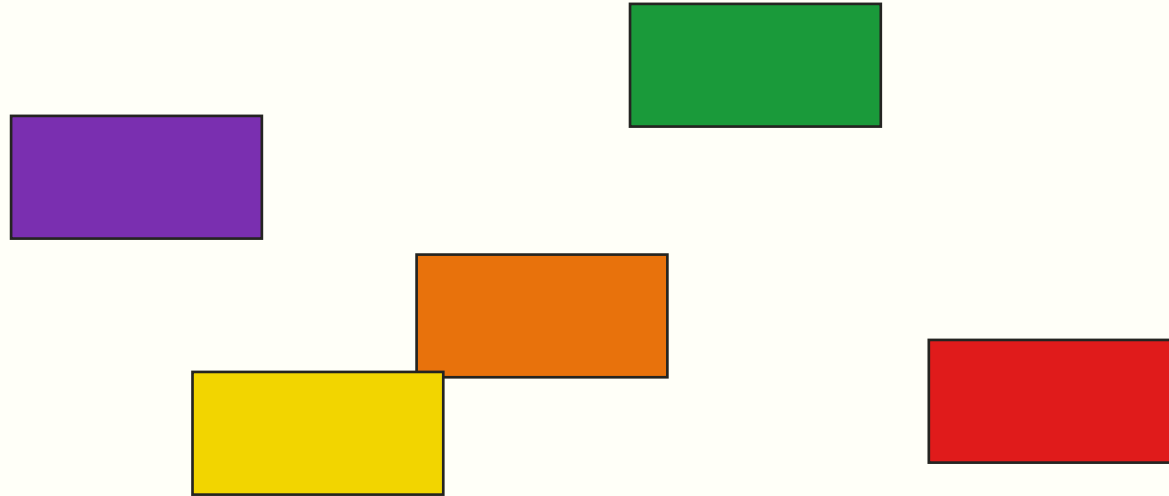


Scale and axis need not be straight lines. For instance polar coordinates are frequently more informative for certain kinds of data.



Interpolation choices provide different perceptions of continuity

Put these five in order from low to high.



Kinds of Color Scales

Quantitative



Ordinal

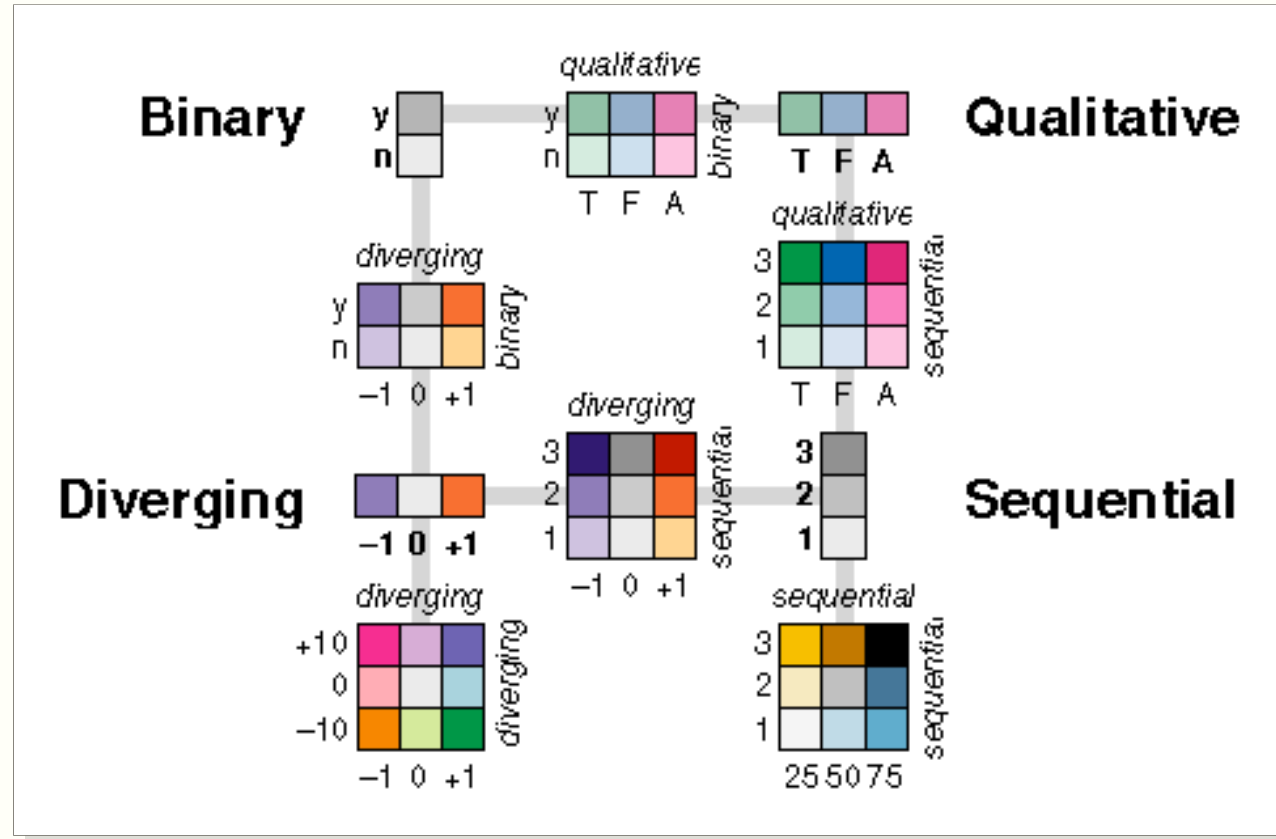


Diverging



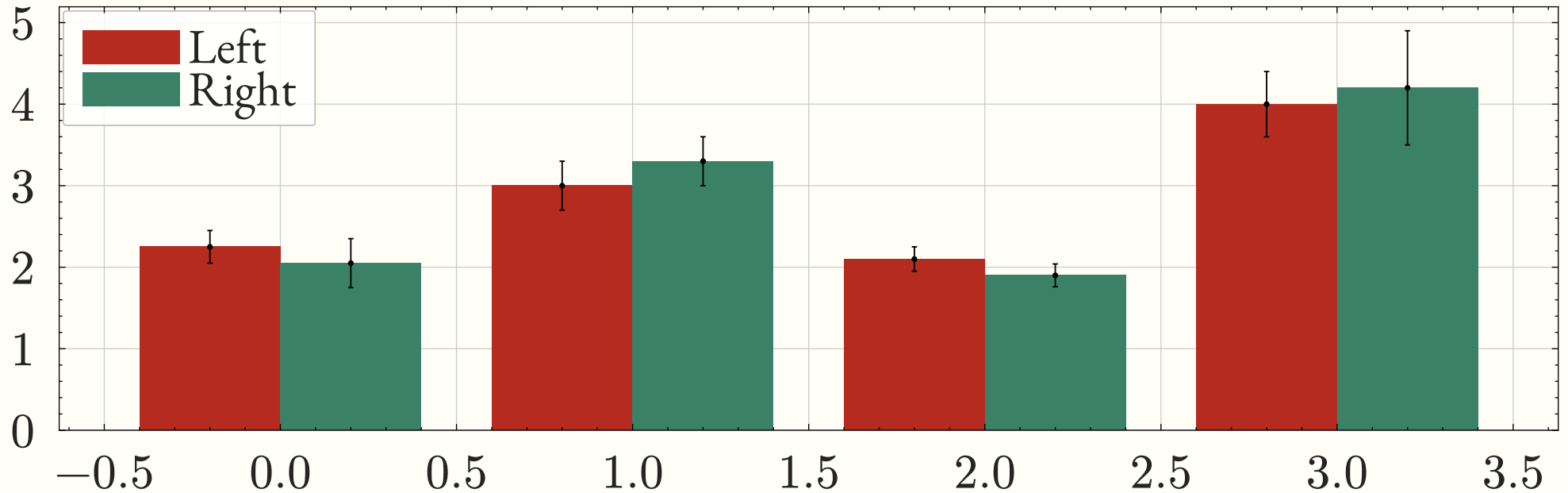
Categorical



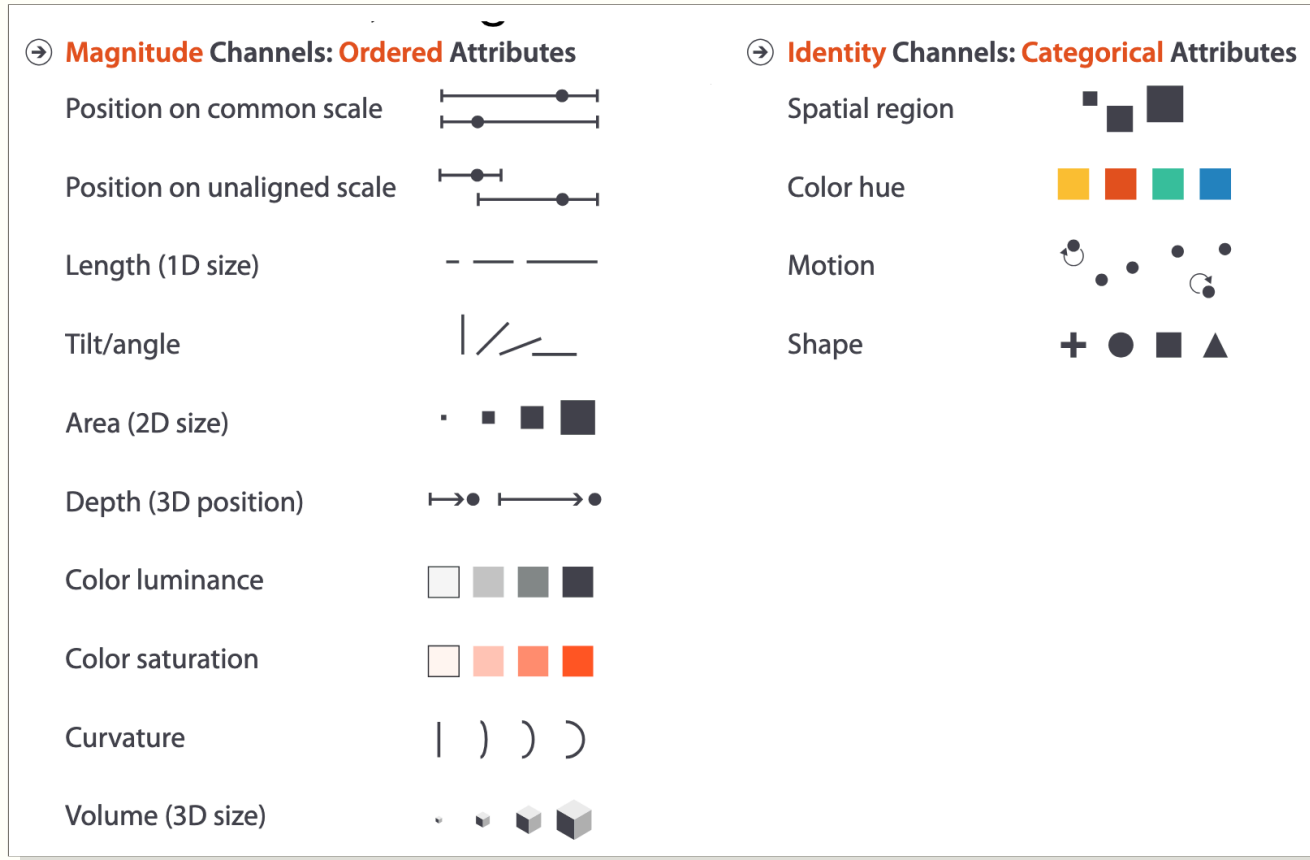


Munzner, Visualization Analysis and Design, 2014.

Remind you of anything?

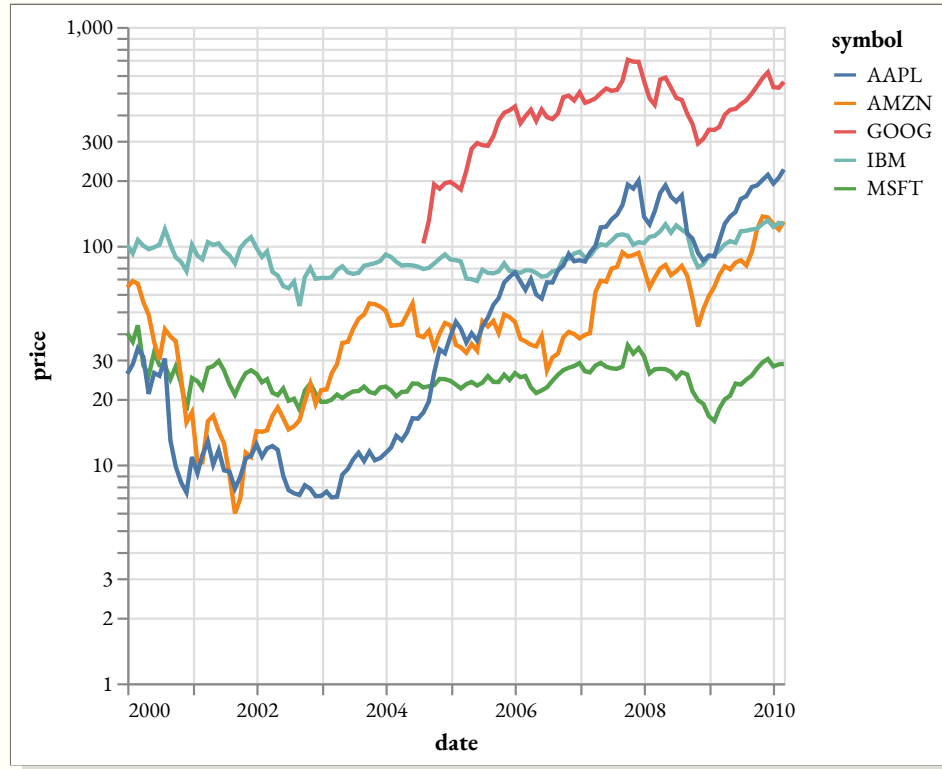


Channels require different scales



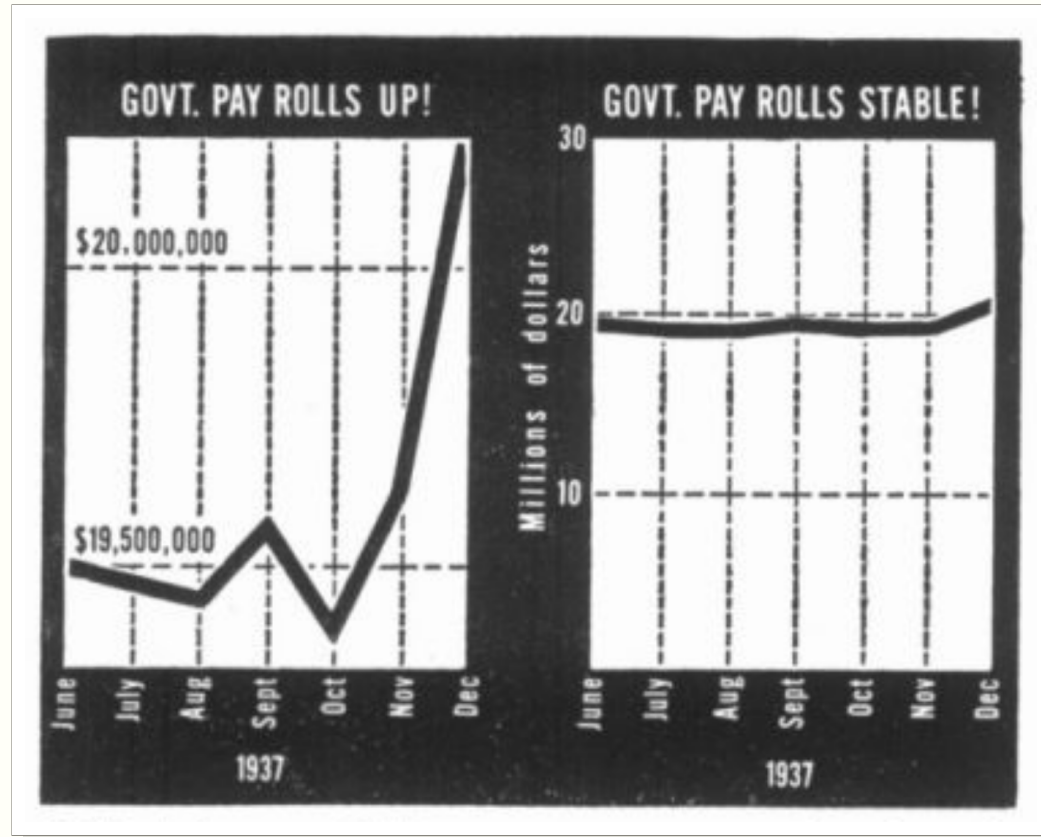
*A **scale** is a function.*

*An **axis** is its **visual** representation.*

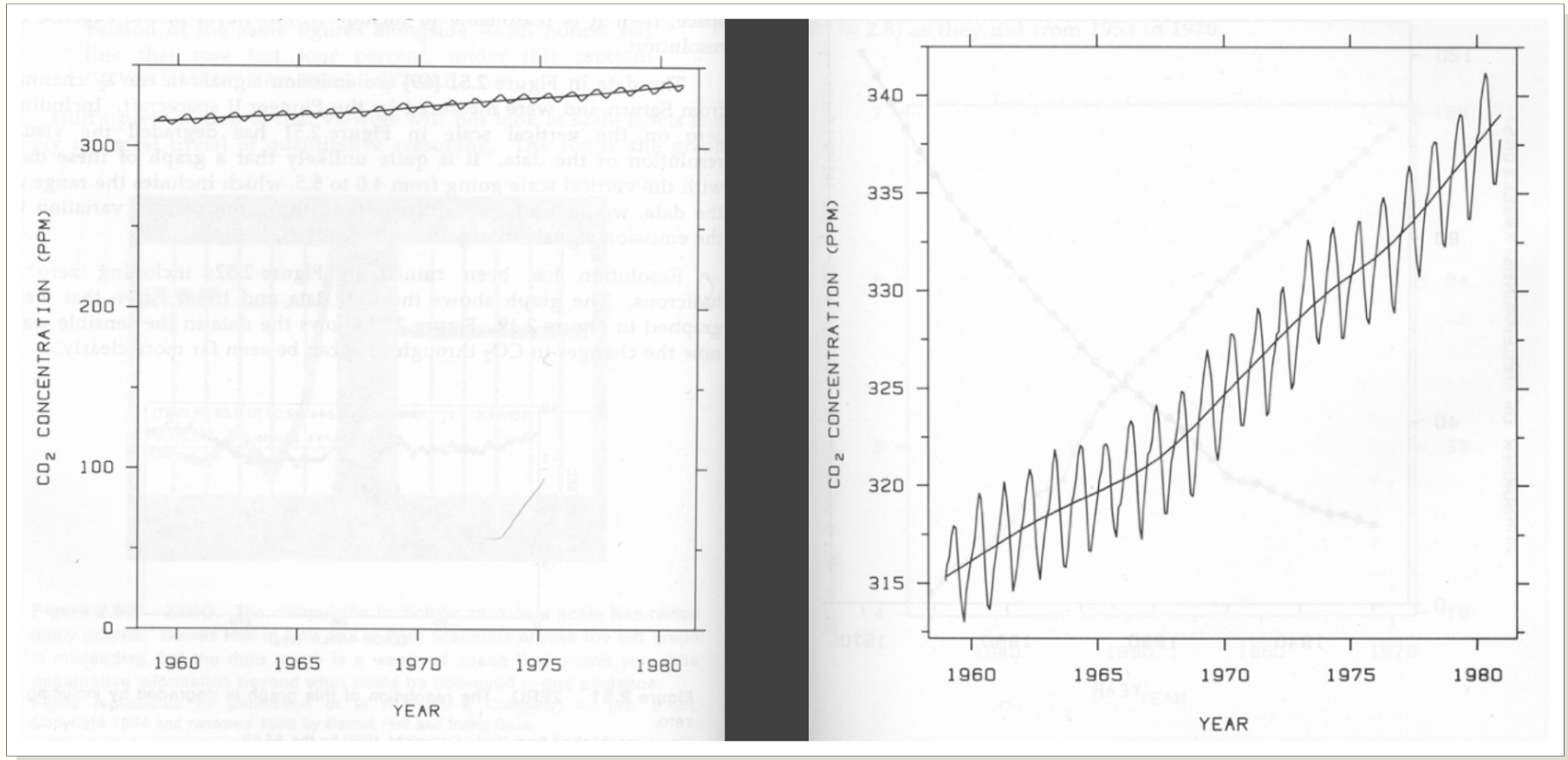


- Ticks can show scale distortions
- Gridlines are extensions of the axis
- Axis labels fix *data units* at the corresponding *visual units*

Include zero?

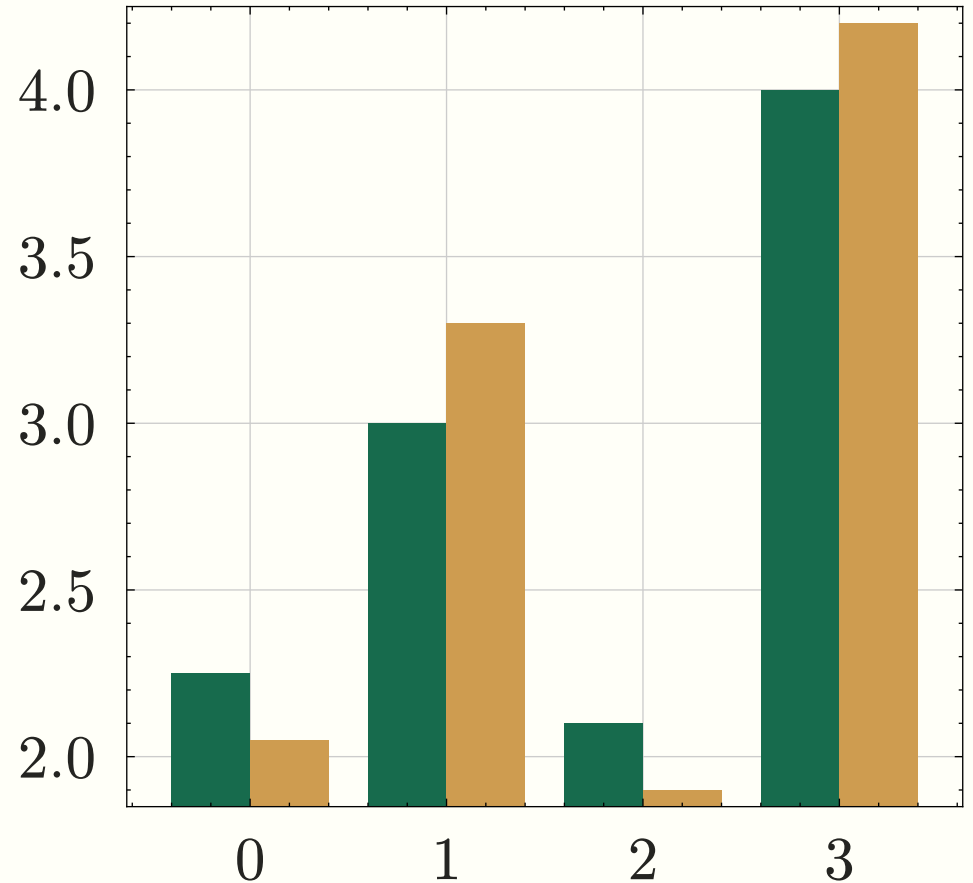
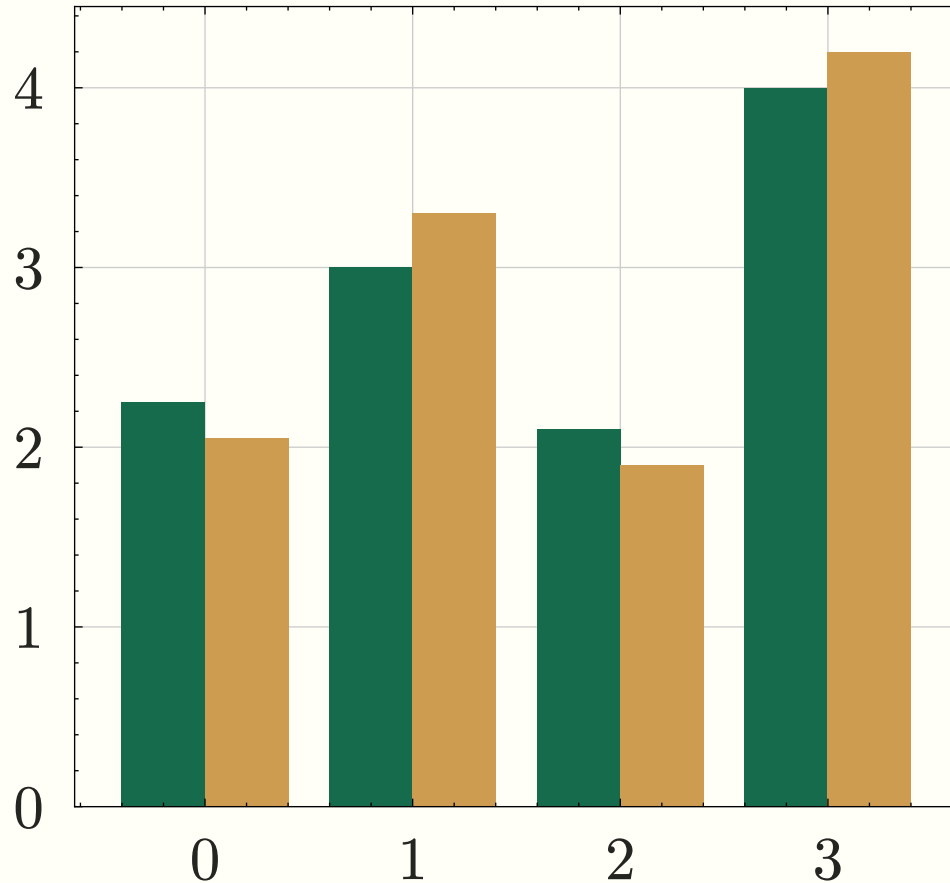


Huff, How to Lie with Statistics, 1954.



Yearly CO₂: a zero baseline hides the trend.

Size axes should almost always starts at zero.

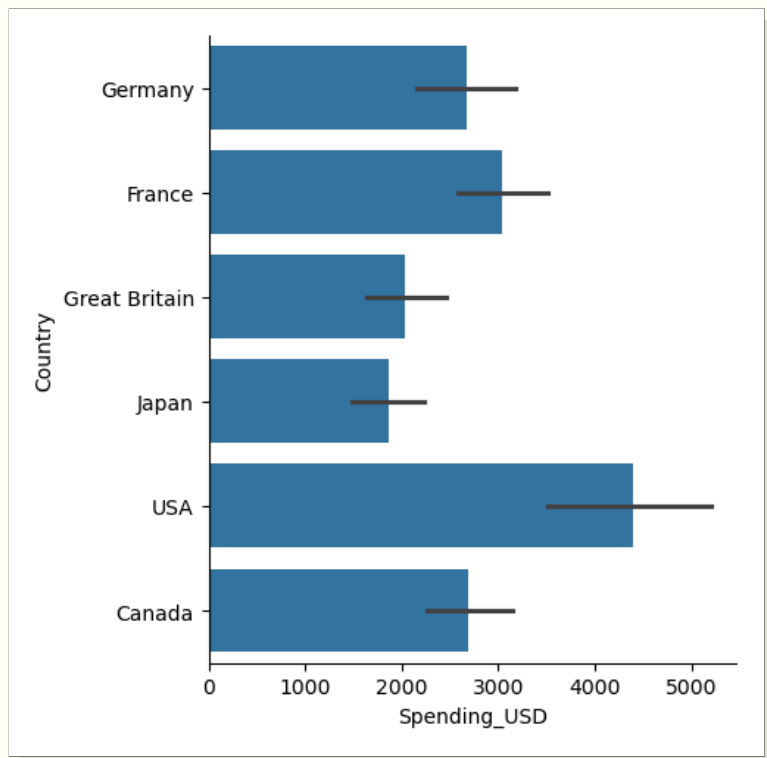


*Once you learn the **grammar** of graphics.*

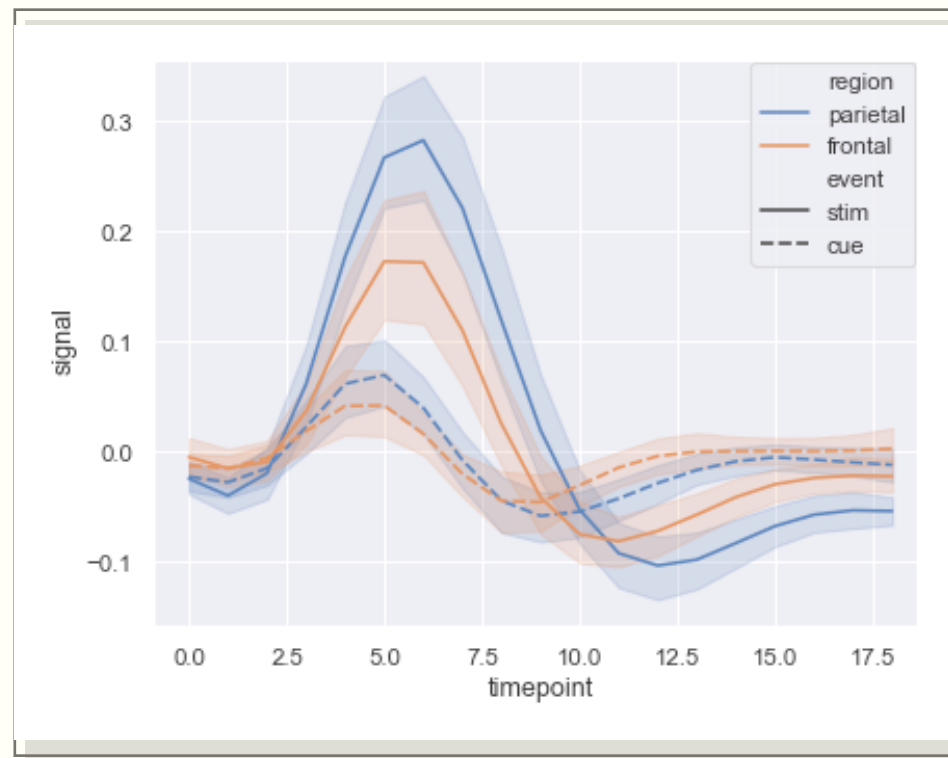
*You still need to develop the skills to **speak the language fluently.***

Showing aggregates and distributions.

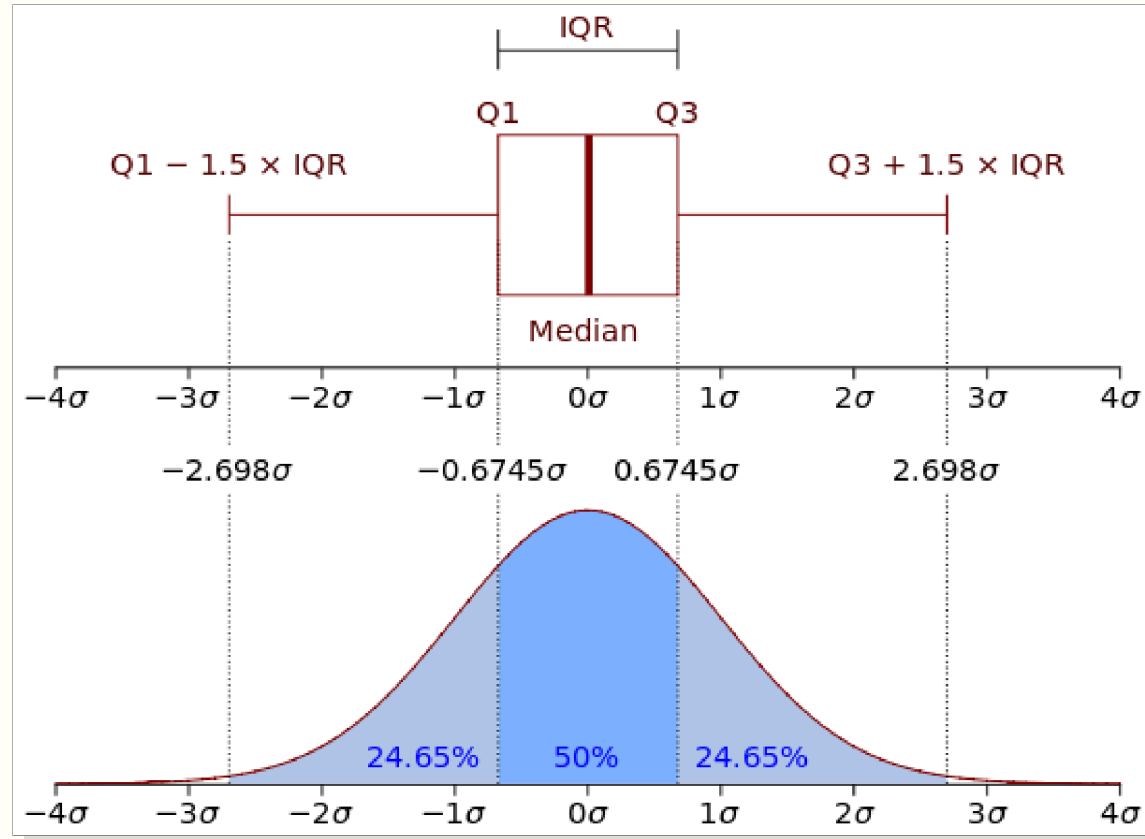
Frequently we need a single mark to represent multiple data items.



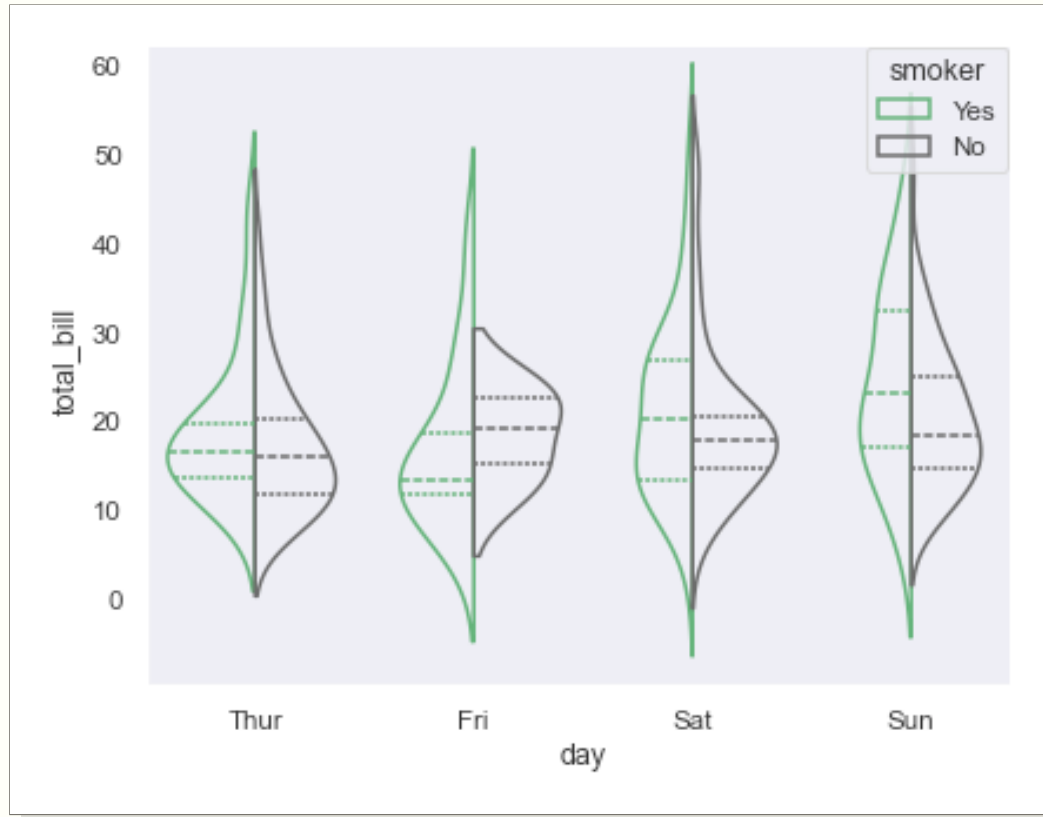
Bar + error bar



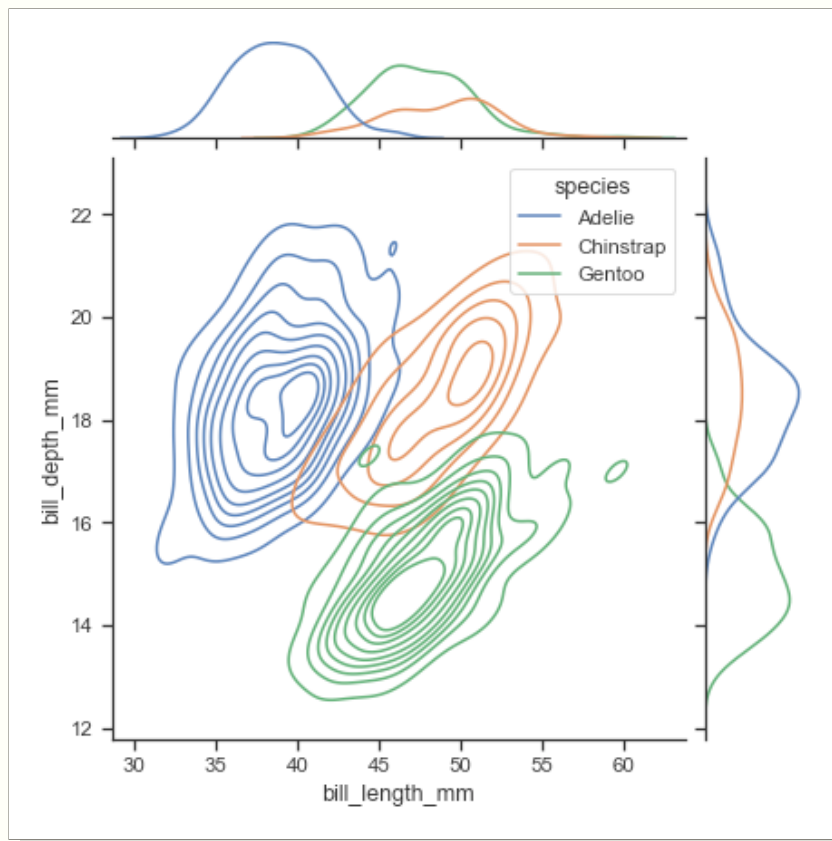
Line + confidence band



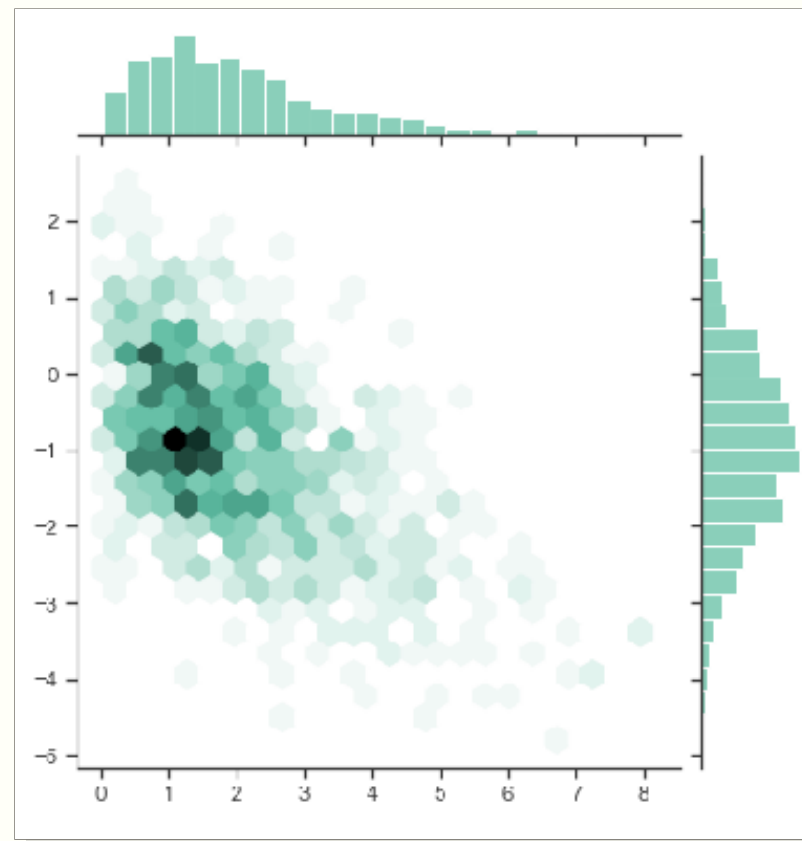
The Tukey box plot: median, quartiles, and whiskers at $1.5 \times IQR$.



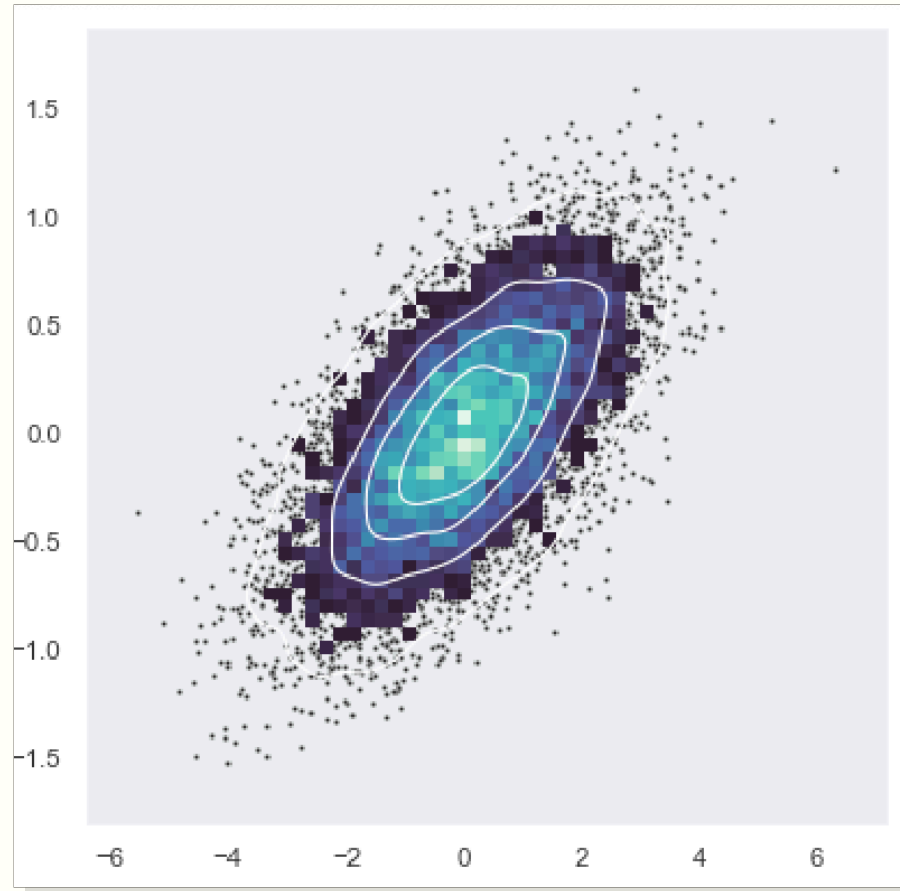
Violin plot: the full density, at the cost of precision.



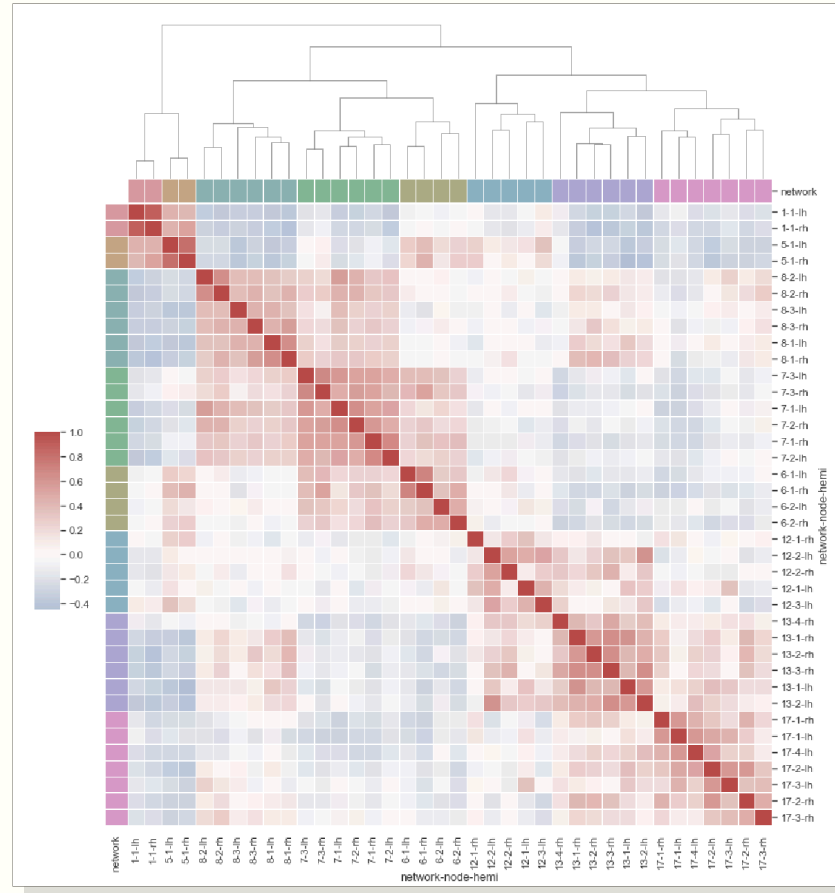
Histogram density



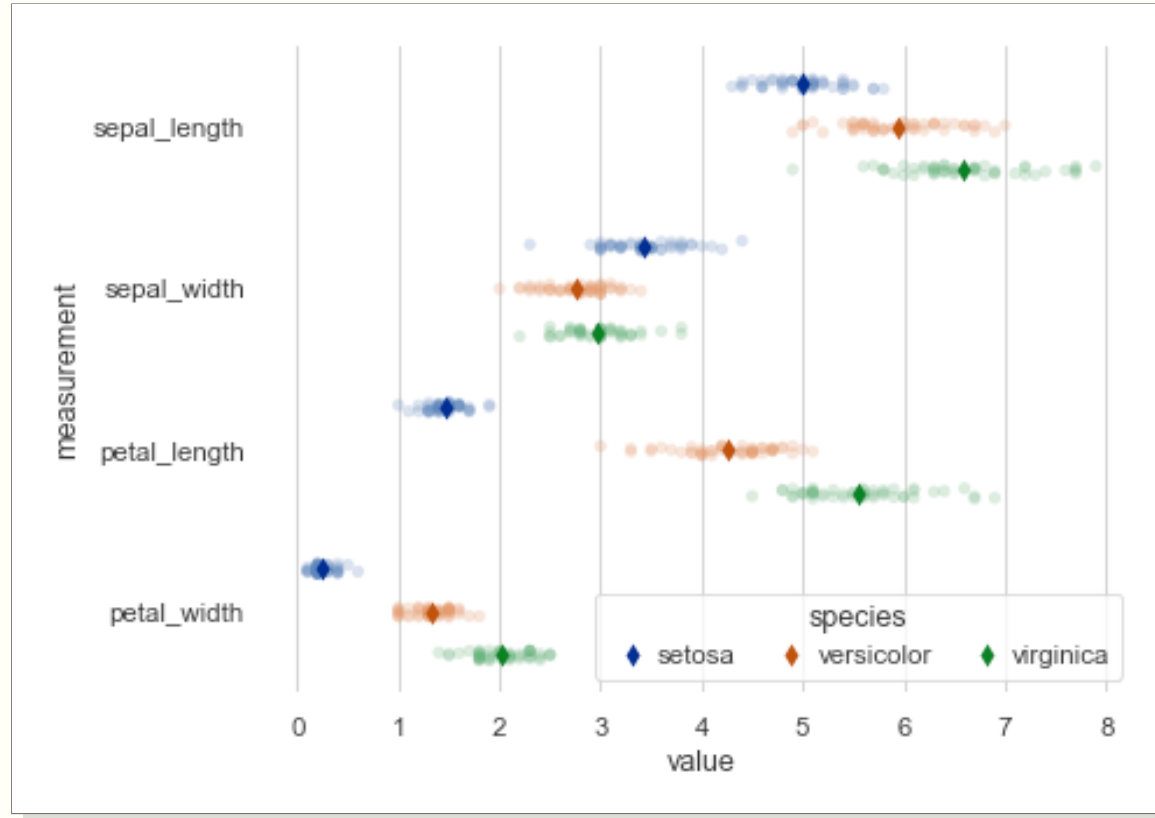
Hex Bin Histogram



Mixed marks density visualization

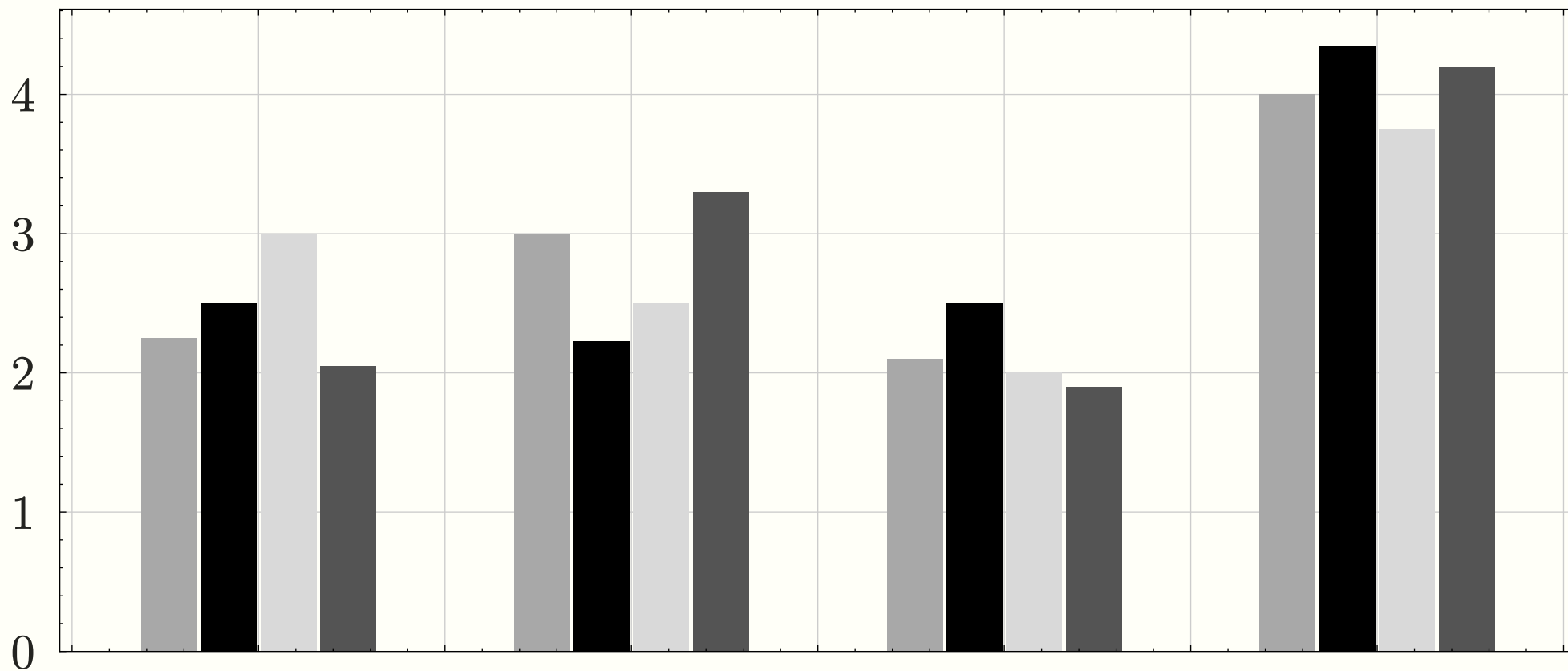


Categorical histogram with hierarchical clusters

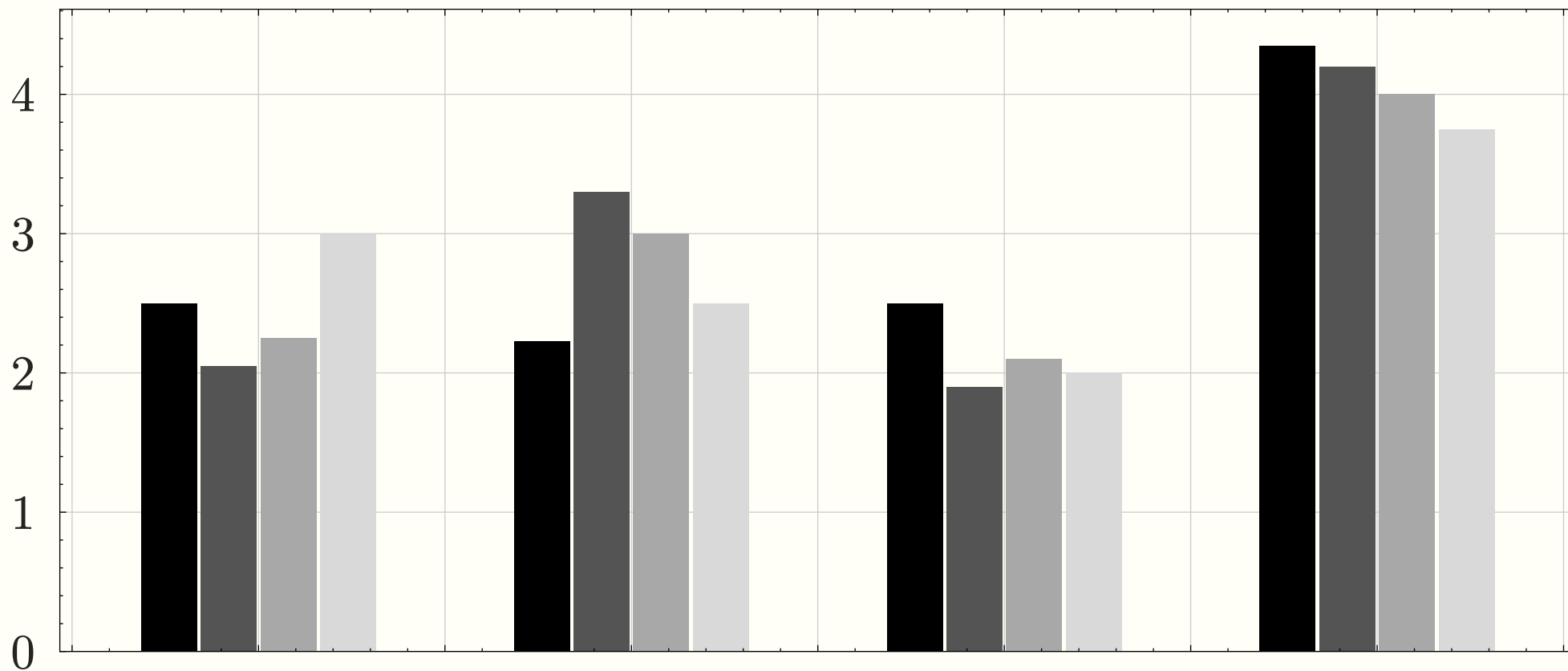


Visual Aggregation: Overplotting transparent, jittered points let density show through a crowded strip.

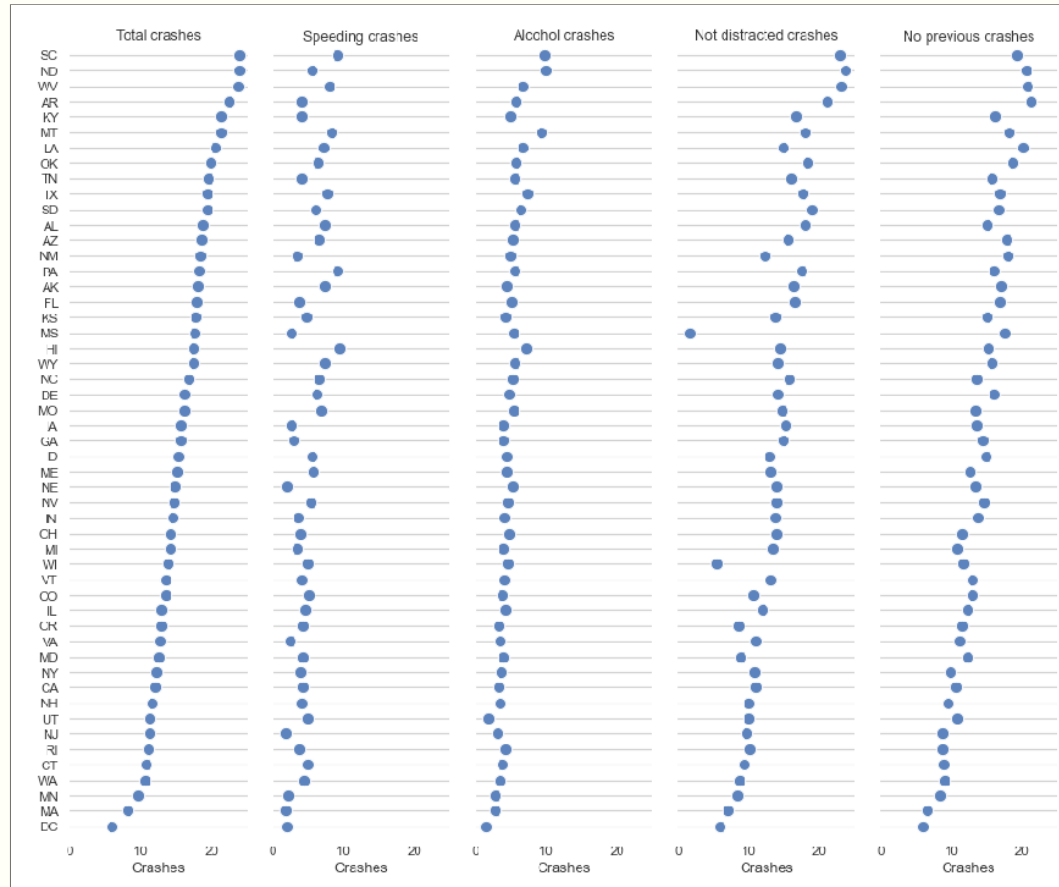
Think about categorical ordering

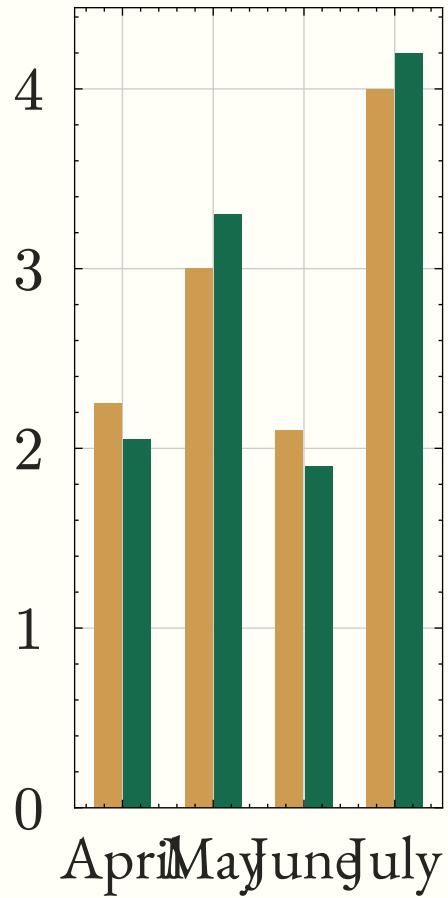


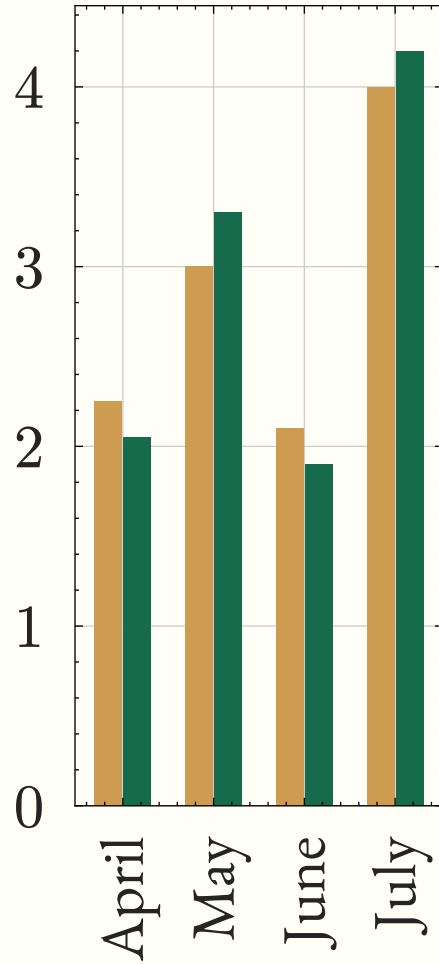
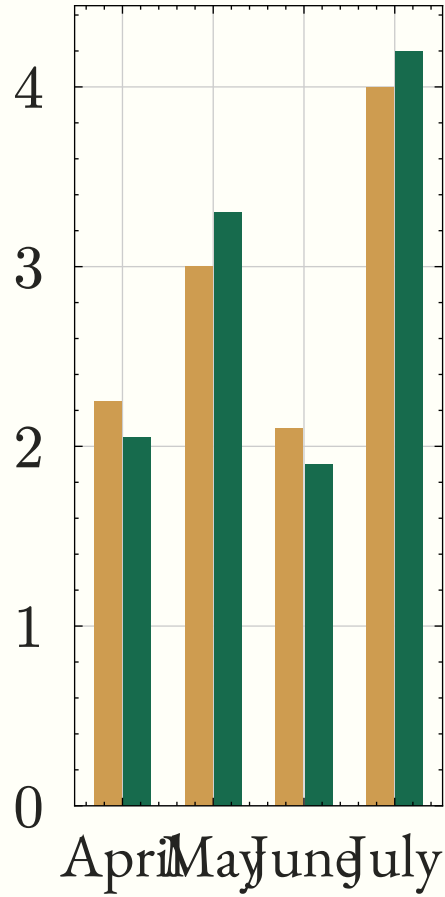
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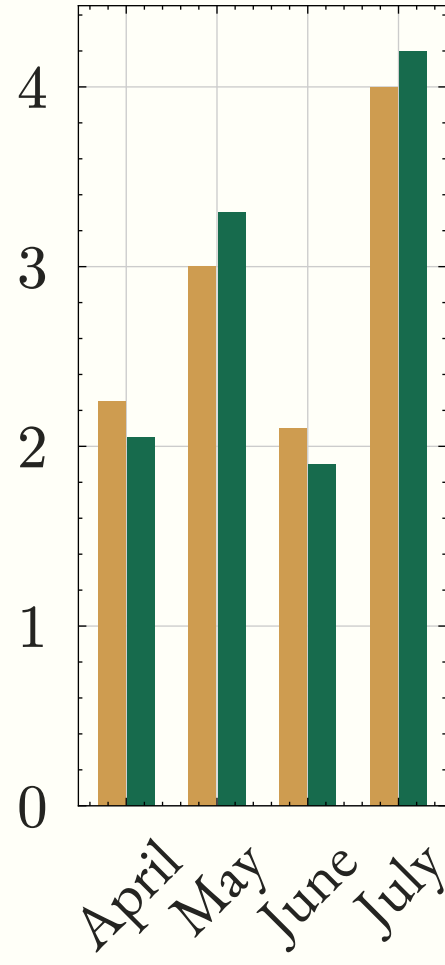
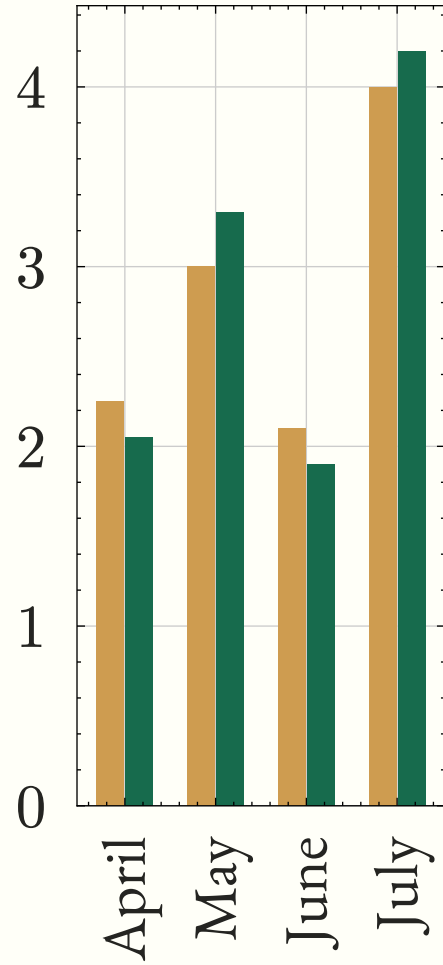
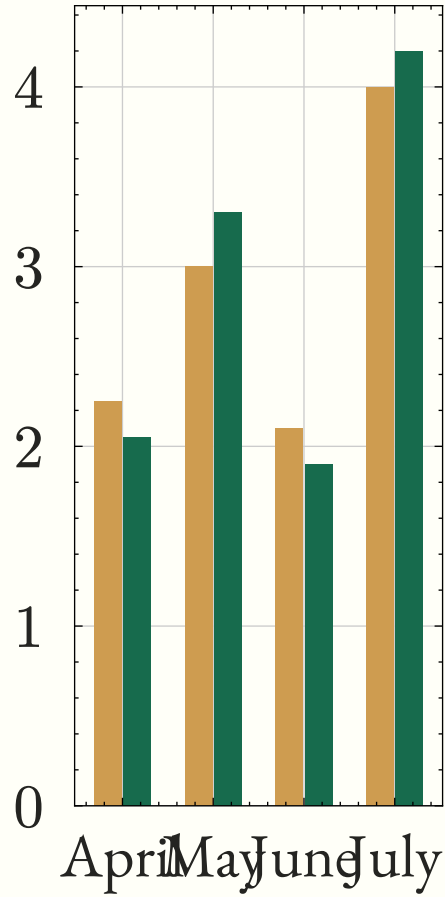


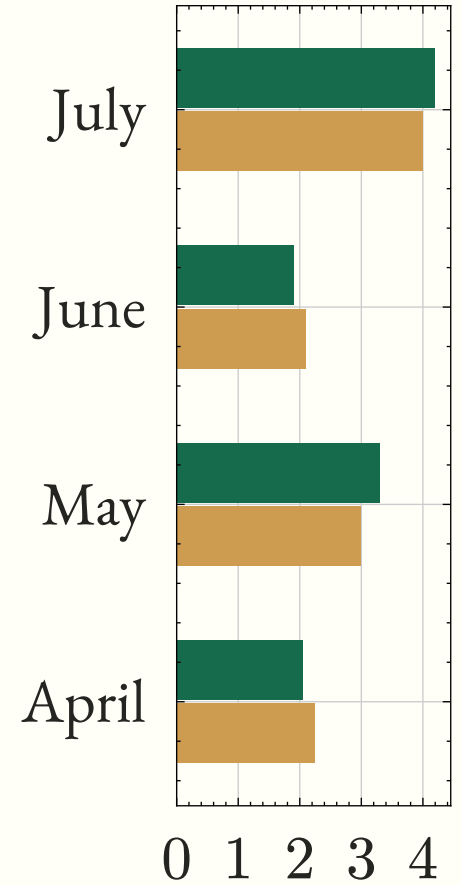
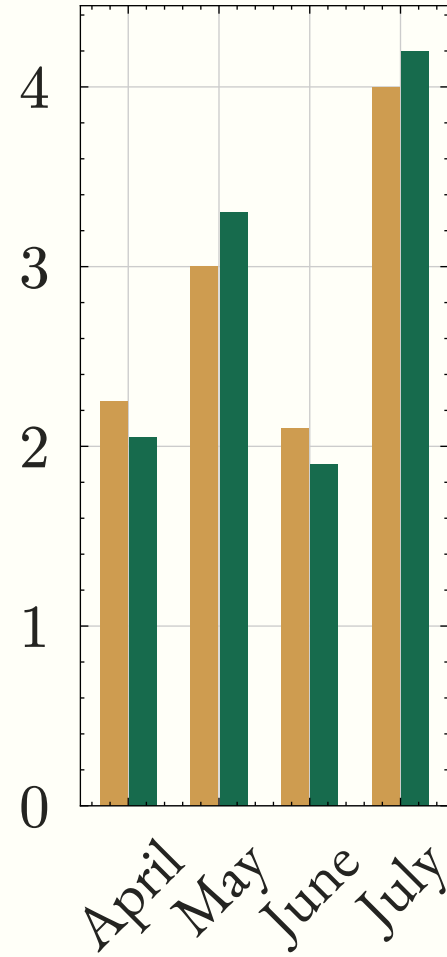
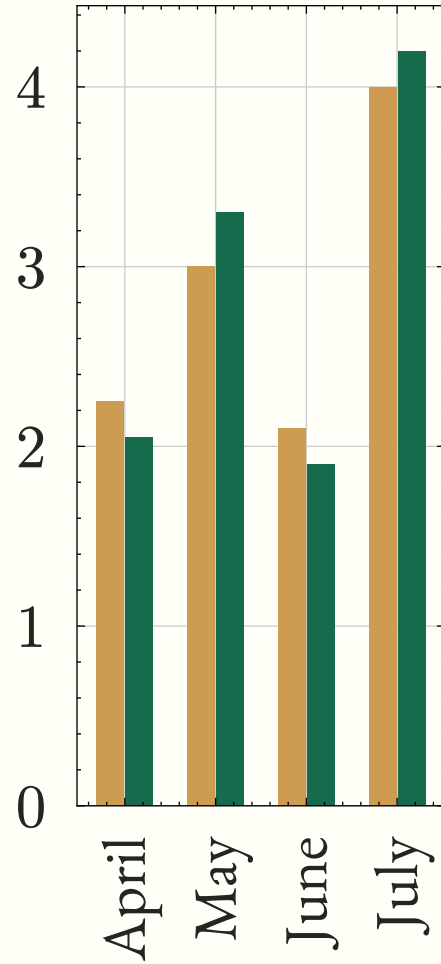
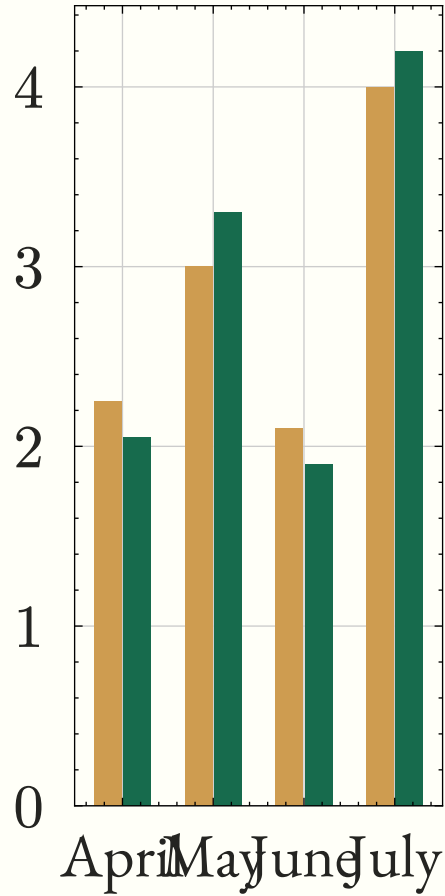
Think about categorical ordering

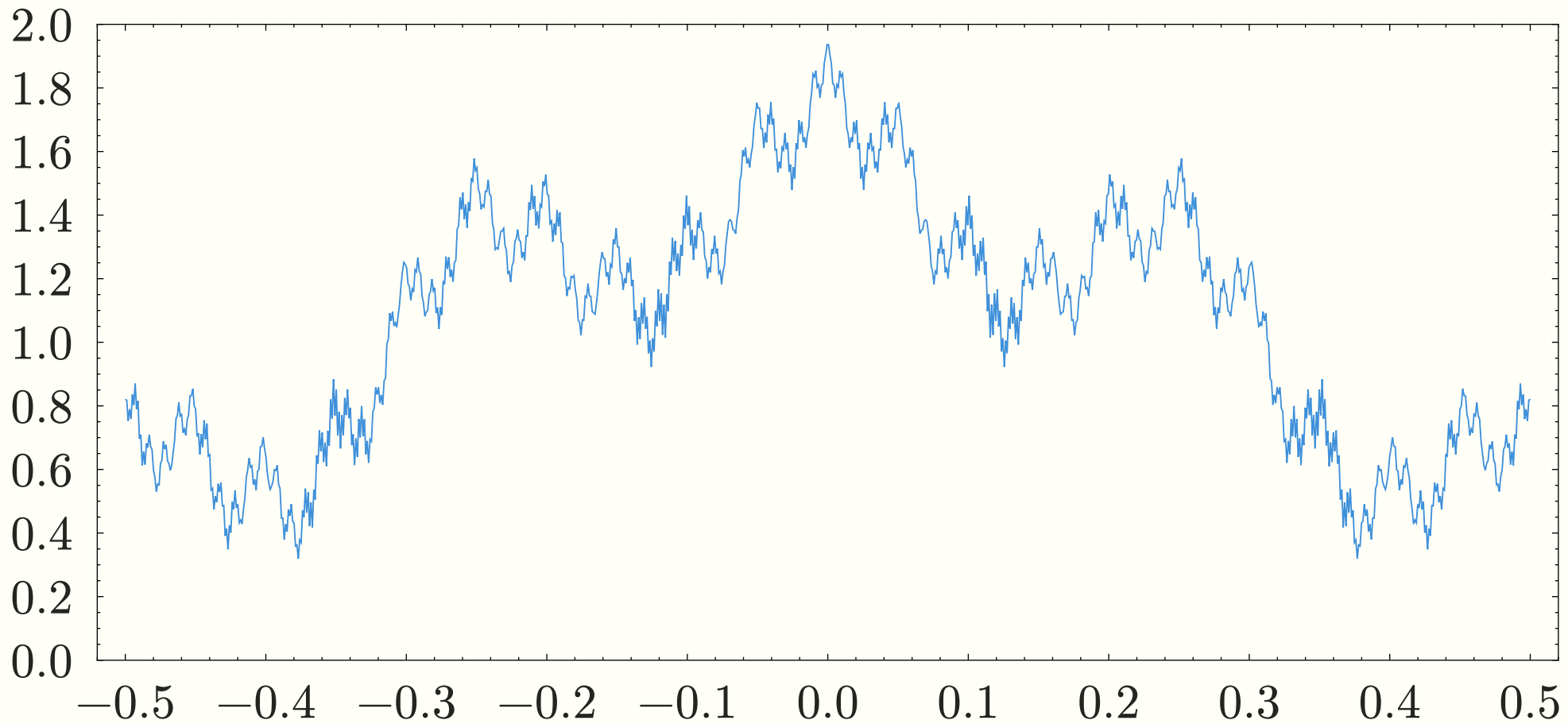


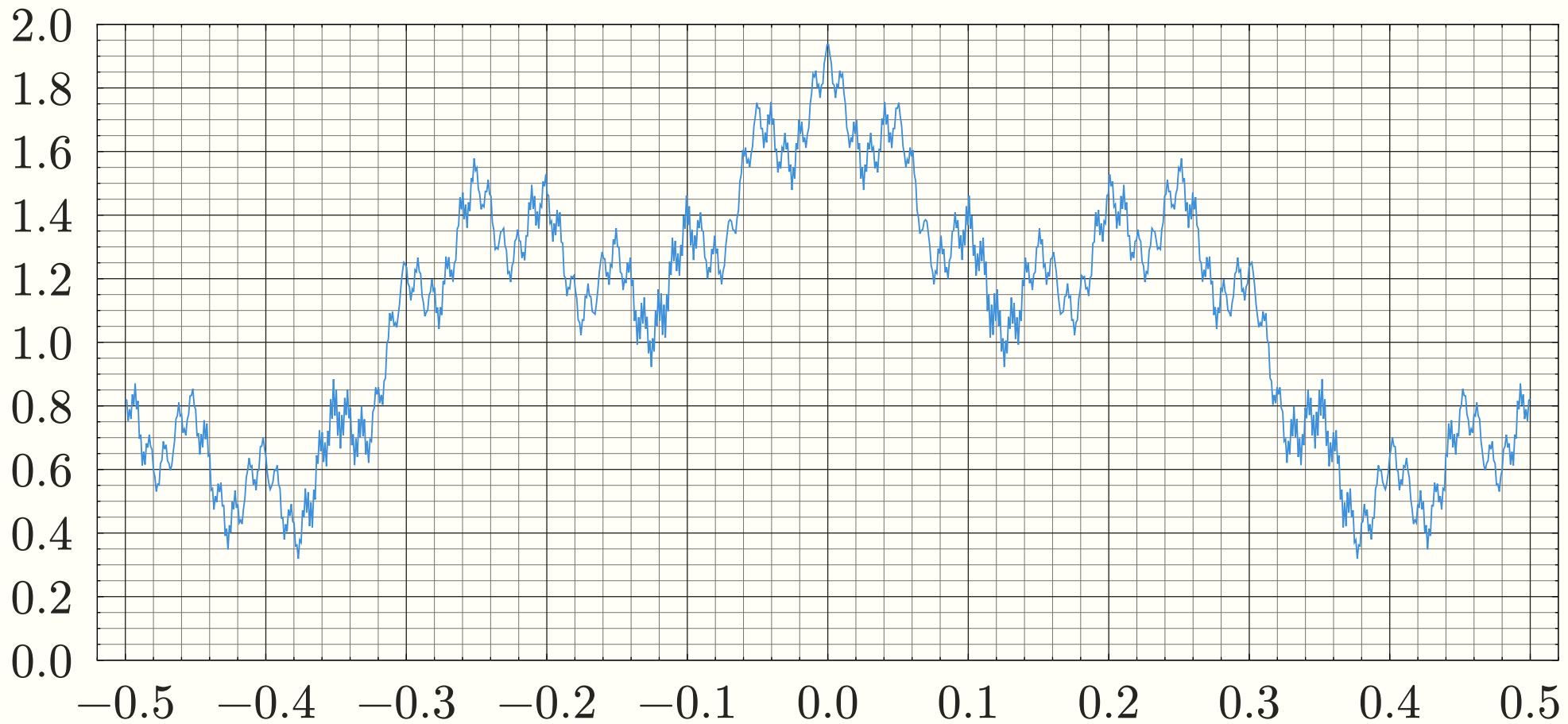


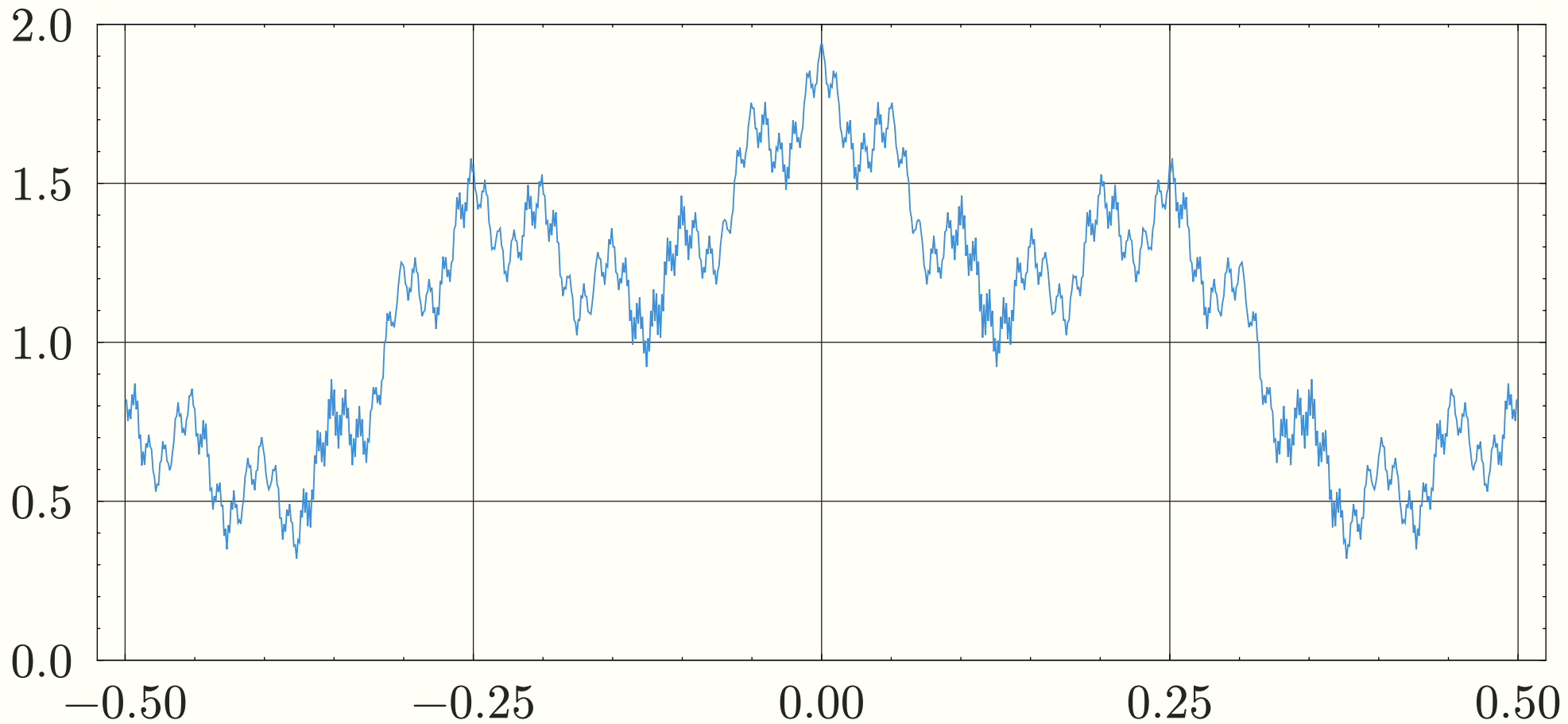


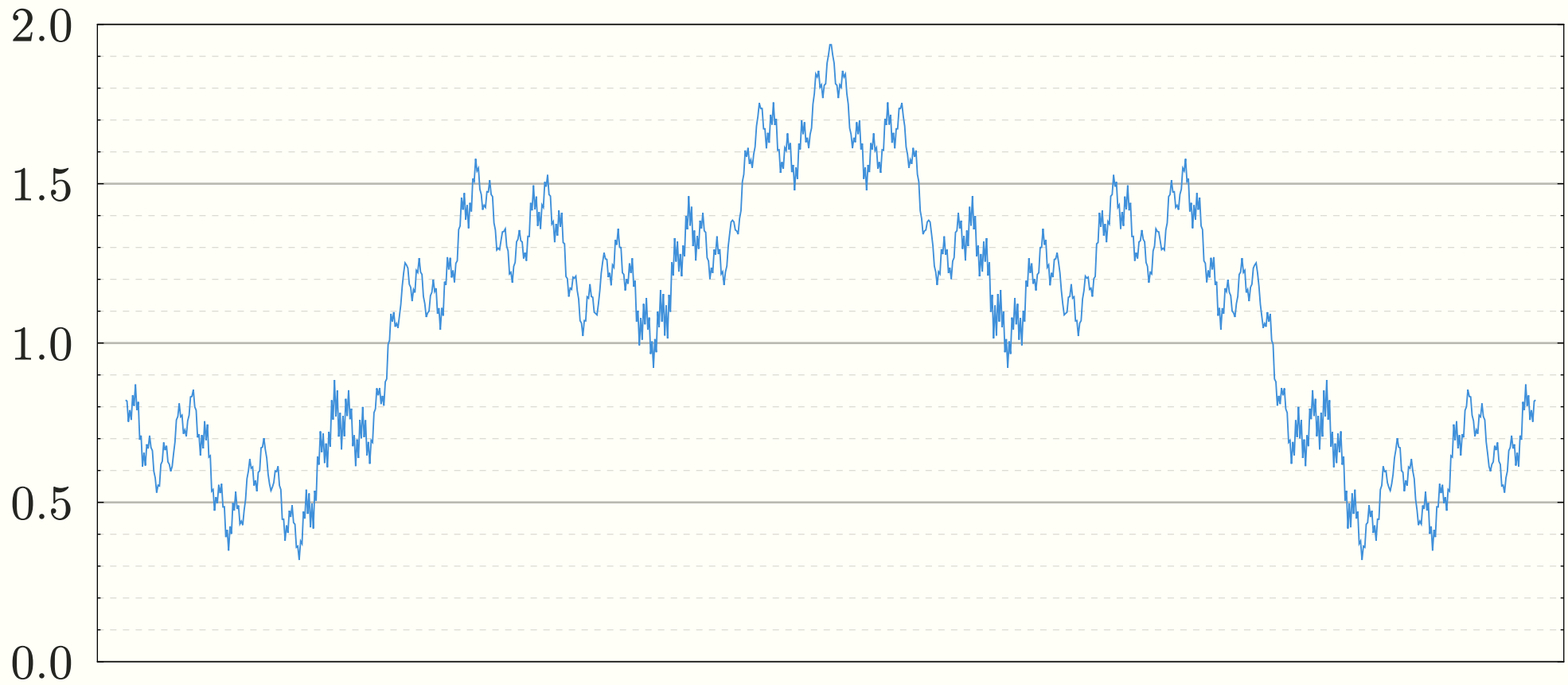


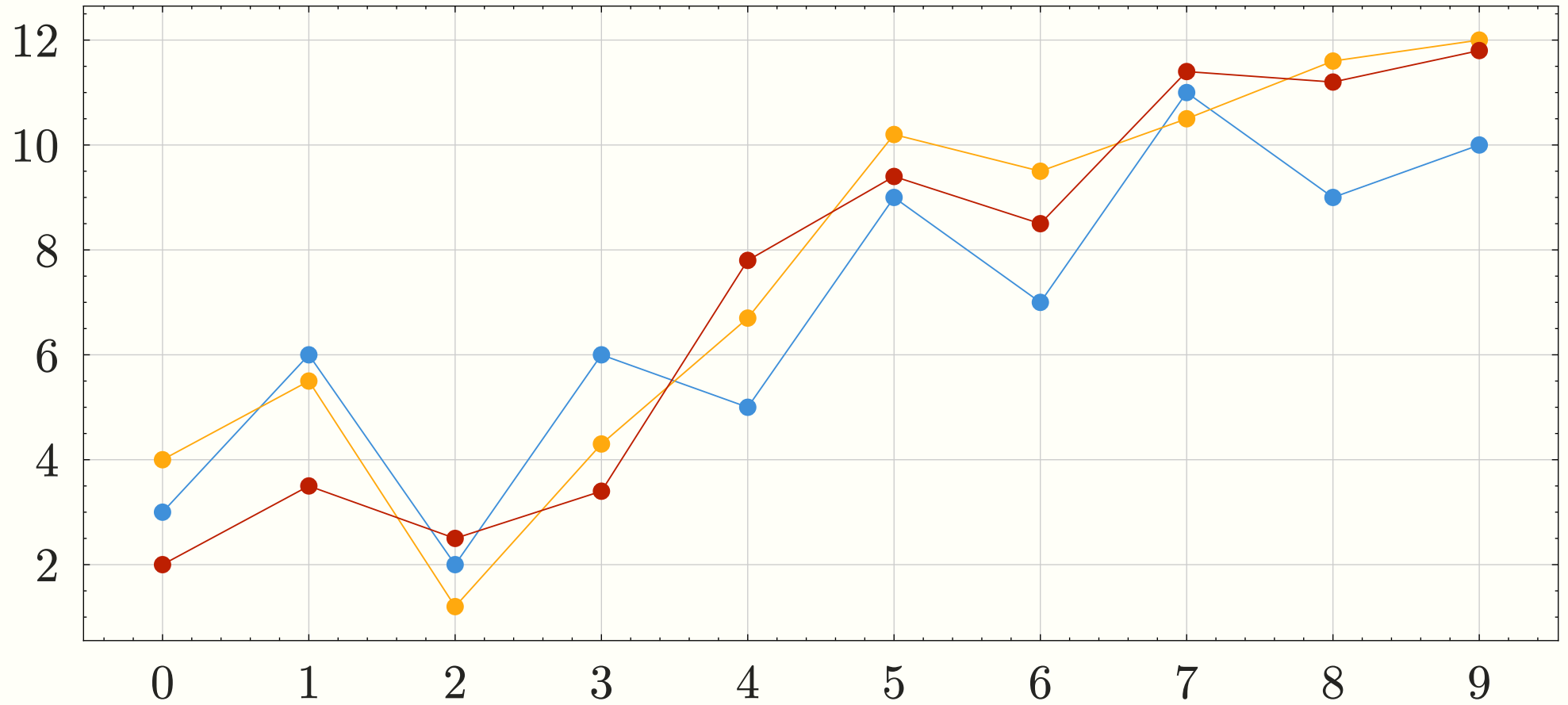


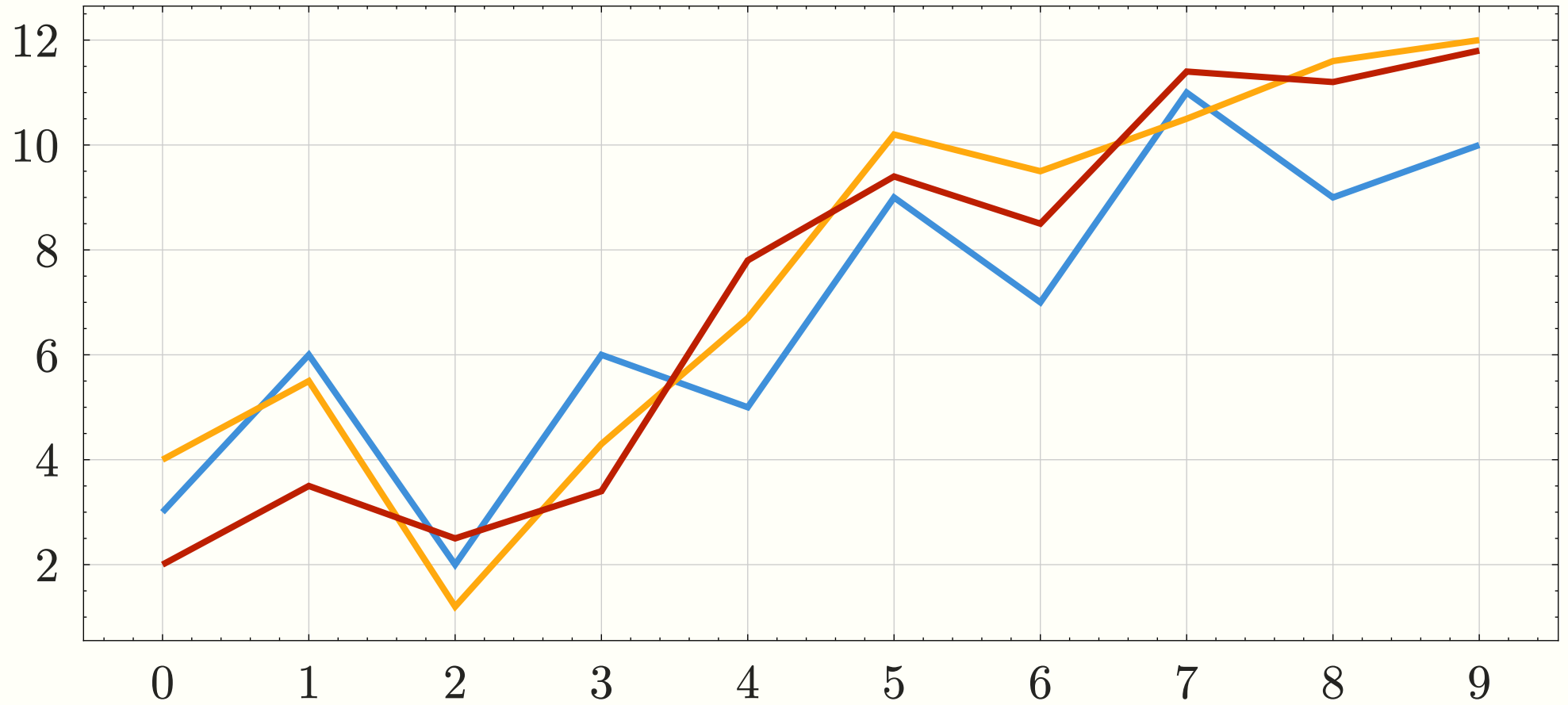


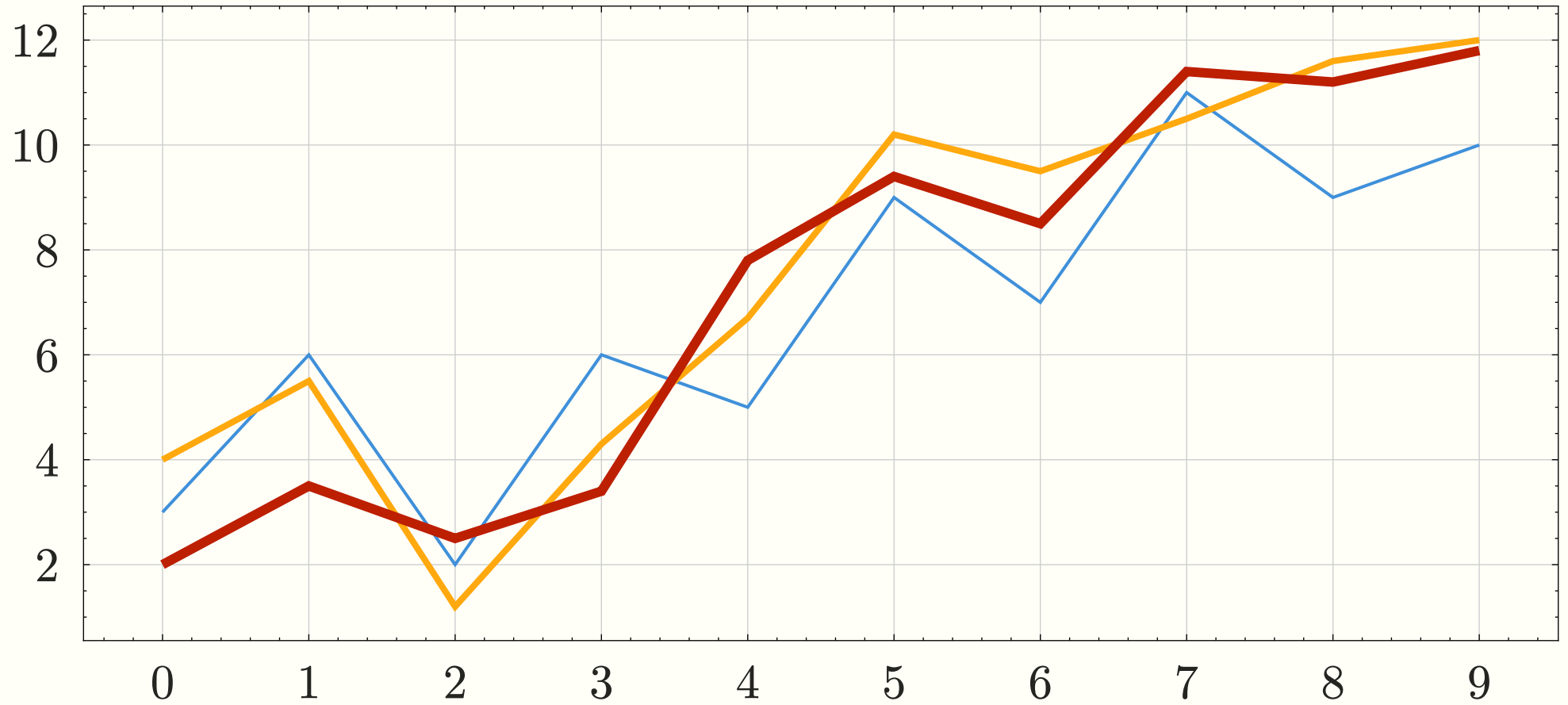


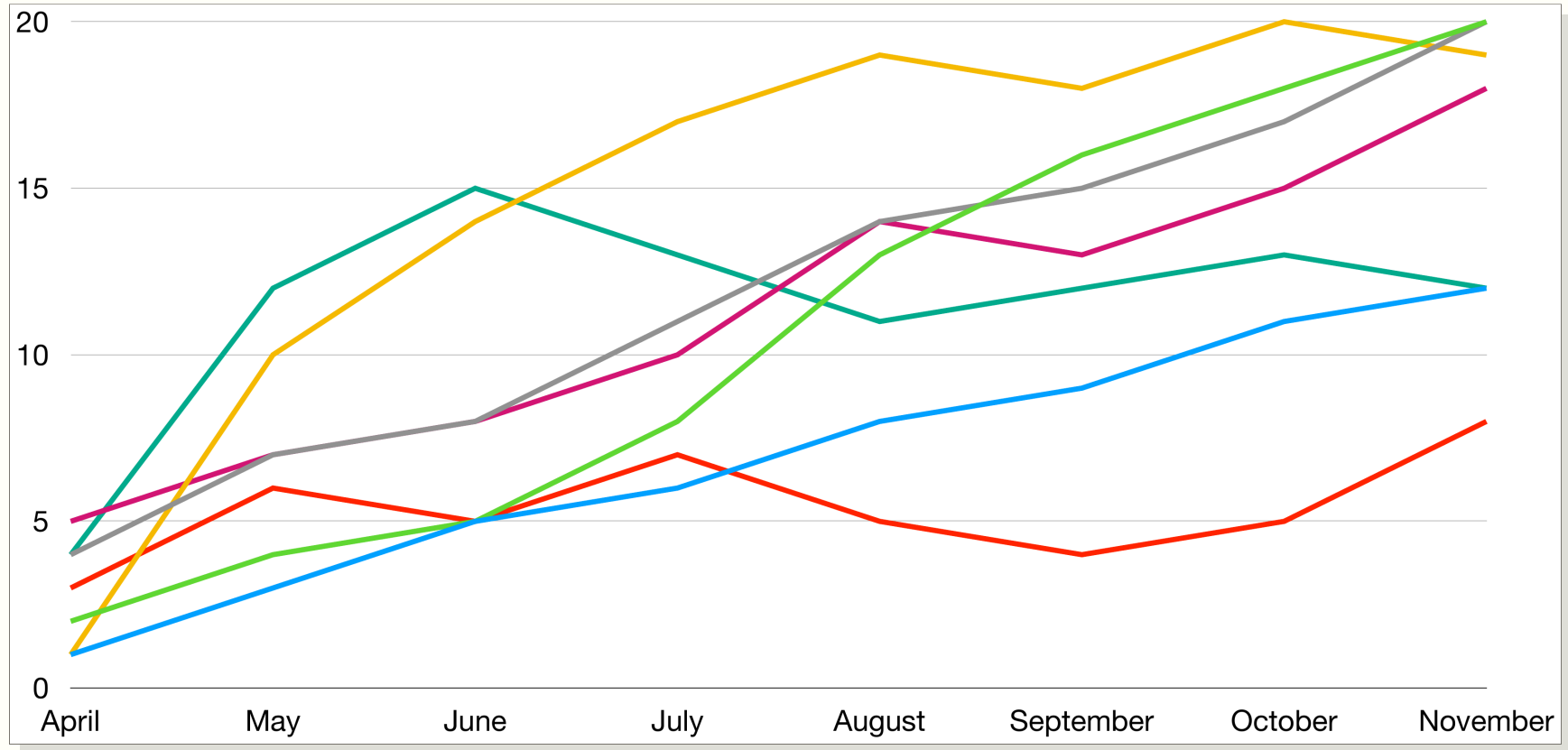


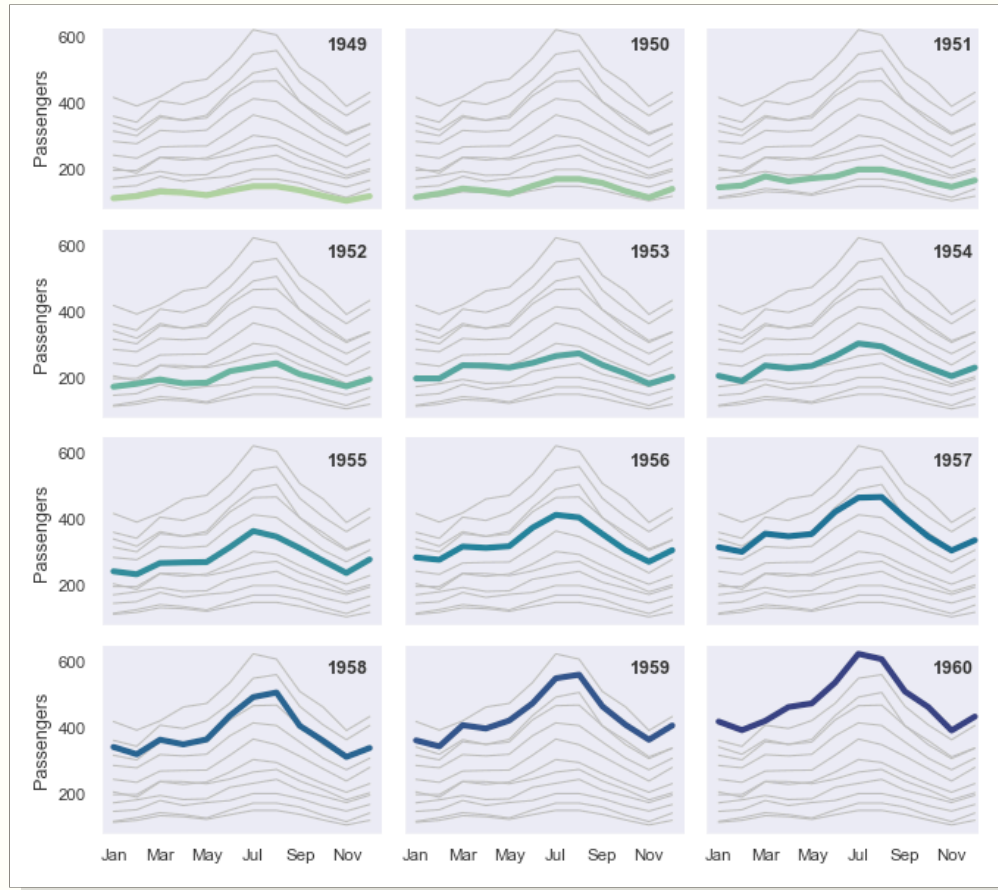


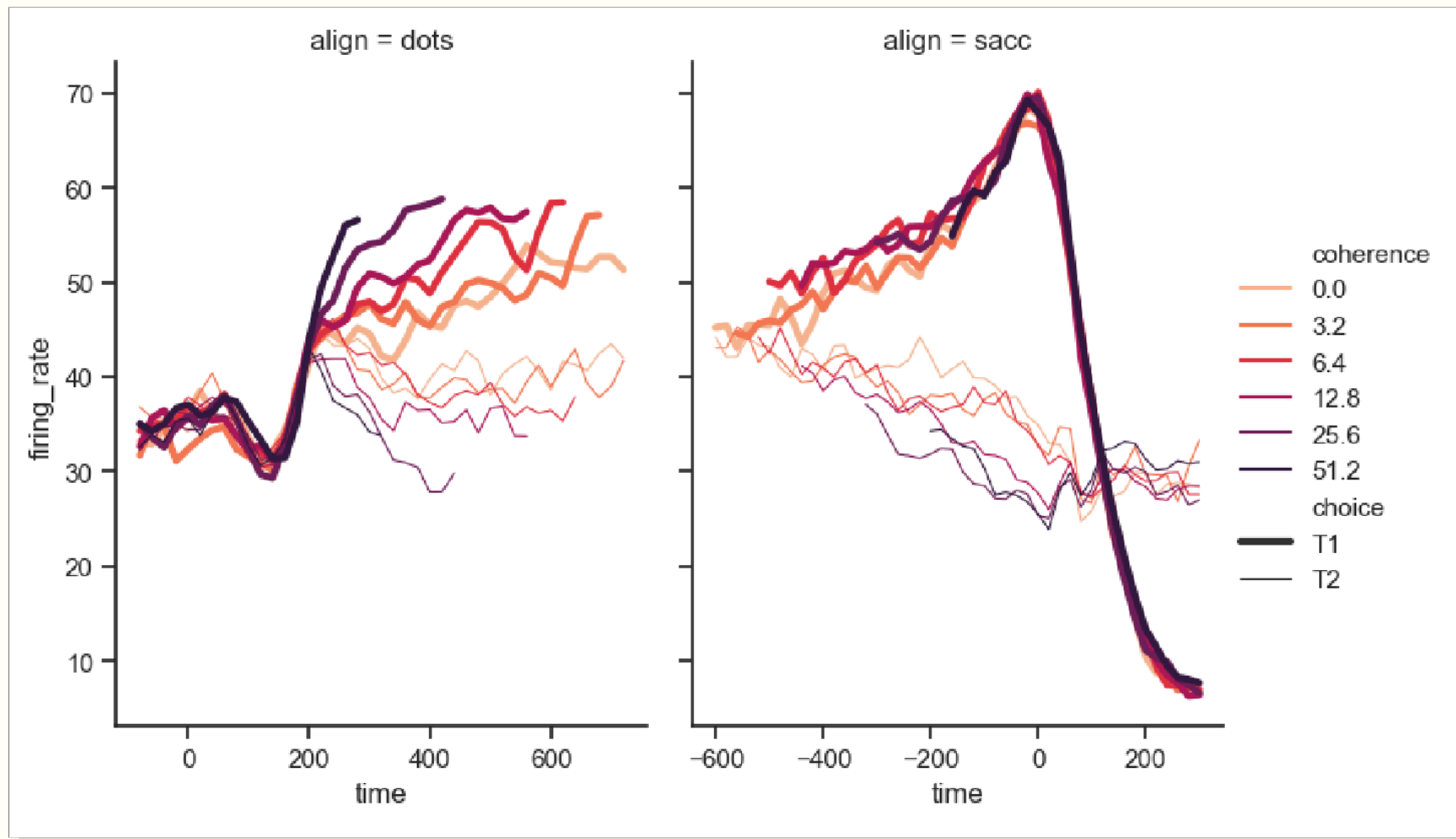


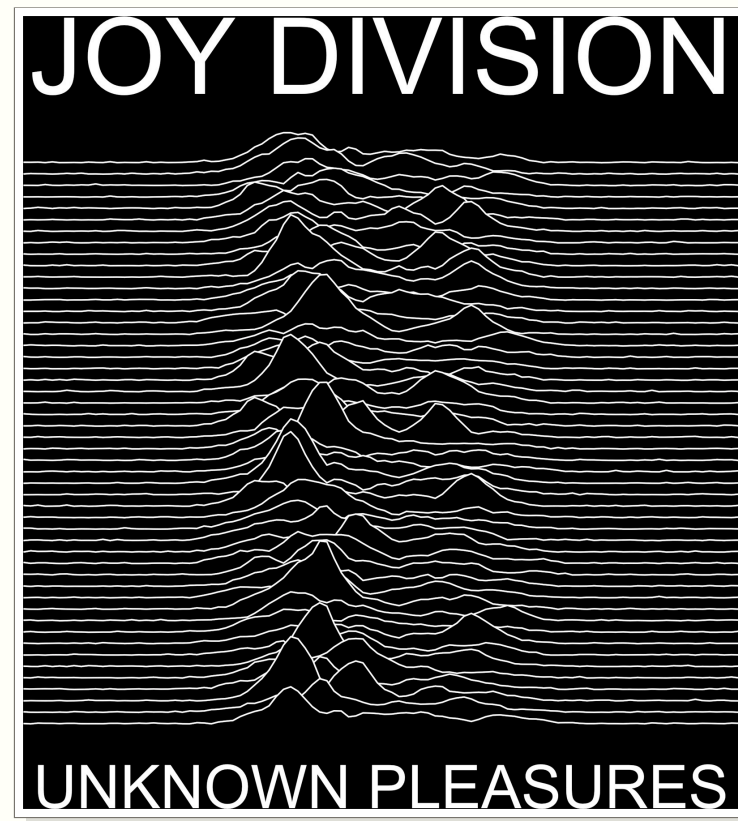
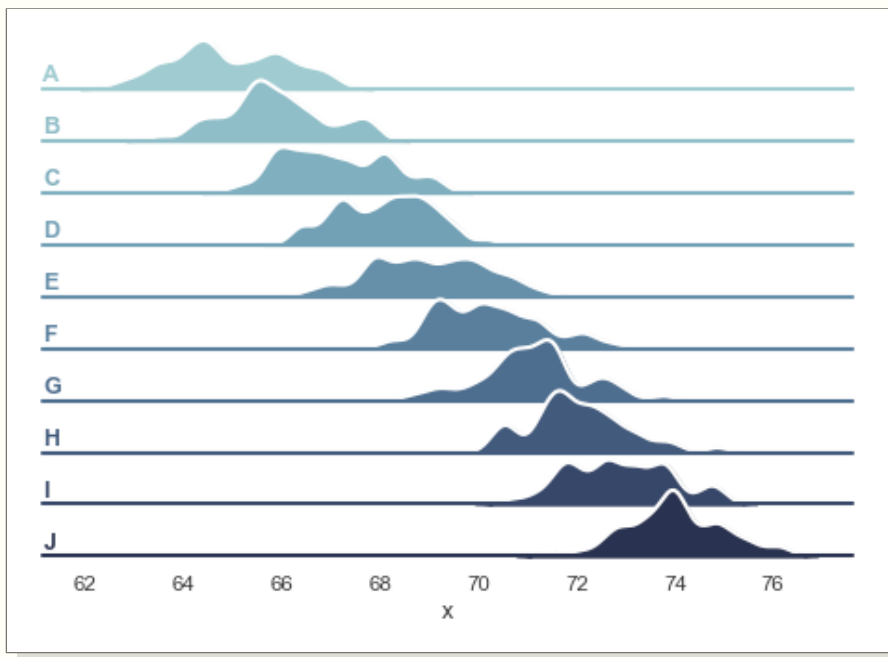












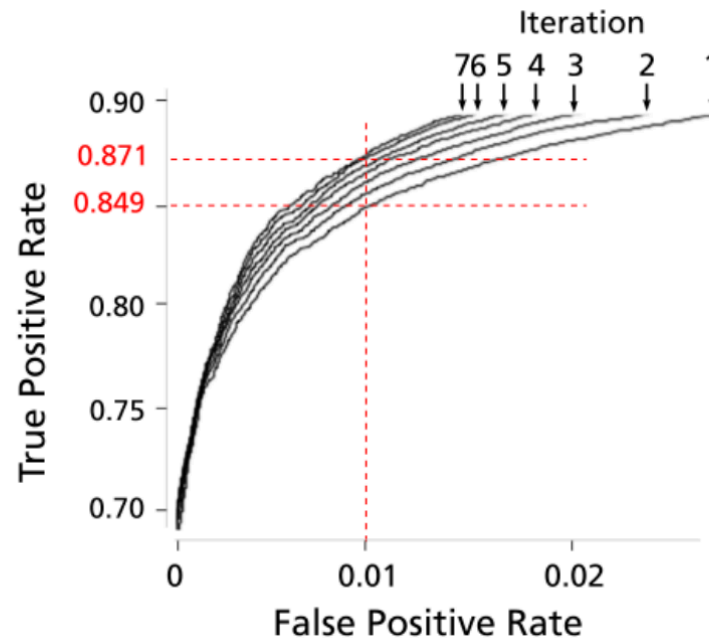


Figure 8: ROC curves of 7 iterations; true positive rate incrementally improves.

Annotate the point that matters: reference lines and a callout beat a caption.

Role	Name	Year of the...	Debut	Number of Fans	Takedown Rate
Face (The Hero)	The Ultimate Warrior	Tiger	May-2011	97320.00	86.2
Face (The Hero)	Hulk Hogan	Oxen	Jan-2008	988551.00	61.978
Face (The Hero)	Macho Man Randy Savage	Monkey	Feb-2008	157618.00	59.29
Face (The Hero)	Hacksaw Jim Duggan	Pig	Mar-2008	30300.00	53.4332
Face (The Hero)	Superfly Jimmy Snuka	Dragon	Mar-2008	12341.00	52.7
Heel (The Bad Guy)	Rowdy Roddy Piper	Rooster	Jun-1968	71645.00	45.4
Heel (The Bad Guy)	The Million Dollar Man Ted DiBiase	Rat	Apr-1975	449342.00	43.7689
Heel (The Bad Guy)	Mr. Perfect Curt Henning	Rat	May-1980	13773.00	38
Heel (The Bad Guy)	Jake the Snake Roberts	Snake	Jul-1975	5609.00	37.99
Jobber (The Unknown)	Brad Smith	Sheep	Aug-2008	1103.00	36.316
Jobber (The Unknown)	Ted Duncan	Sheep	Aug-2008	200.00	33.61
Jobber (The Unknown)	Joey the Uber Nerd Cherdarchuk	Snake	Aug-2008	5.00	21.0196

Role	Name	Year of the...	Debut	Thousands of Fans	Takedown Rate
Face (The Hero)	The Ultimate Warrior	Tiger	May-2011	97.3	86.2
	Hulk Hogan	Oxen	Jan-2008	988.6	62.0
	Macho Man Randy Savage	Monkey	Feb-2008	157.6	59.3
	Hacksaw Jim Duggan	Pig	Mar-2008	30.3	53.4
	Superfly Jimmy Snuka	Dragon	Mar-2008	12.3	52.7
Heel (The Bad Guy)	Rowdy Roddy Piper	Rooster	Jun-1968	71.6	45.4
	The Million Dollar Man Ted DiBiase	Rat	Apr-1975	449.3	43.8
	Mr. Perfect Curt Henning	Rat	May-1980	13.8	38.0
	Jake the Snake Roberts	Snake	Jul-1975	5.6	38.0
Jobber (The Unknown)	Brad Smith	Sheep	Aug-2008	1.1	36.3
	Ted Duncan	Sheep	Aug-2008	0.2	33.6
	Joey the Uber Nerd Cherdarchuk	Snake	Aug-2008	0.0	21.0

A table is a visualization

- Right-align numbers
- round to what matters
- drop the heavy rules
- group the rows.

For the first time in at least 80 years, voters associate the Democrats more with sociocultural issues than with class and economic solidarity

Main reasons people say they like the Democratic party (% of all reasons given)



Source: FT analysis of American National Election Studies, based on Party Images in the American Electorate (Brewer, 2008)
FT graphic: John Burn-Murdoch / @jburnmurdoch
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John Burn-Murdoch / Financial Times; annotated by Evan Peck.

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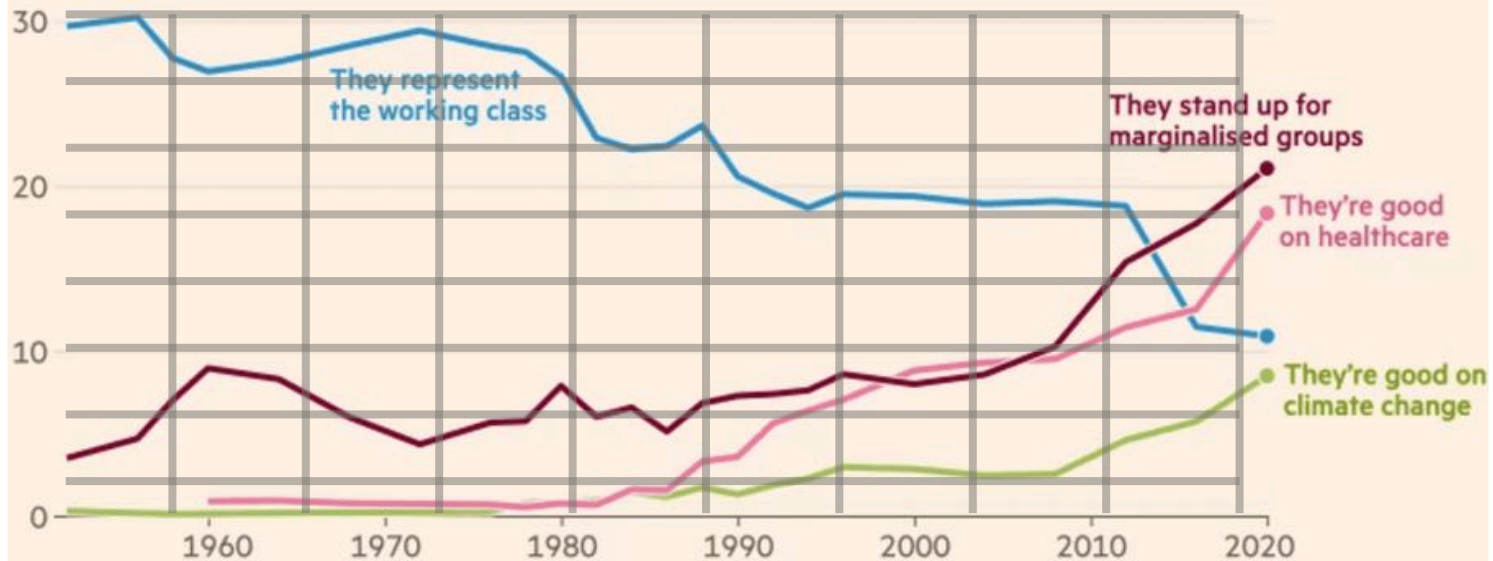


Source: FT analysis of American National Election Studies, based on Party Images in the American Electorate (Brewer, 2008)
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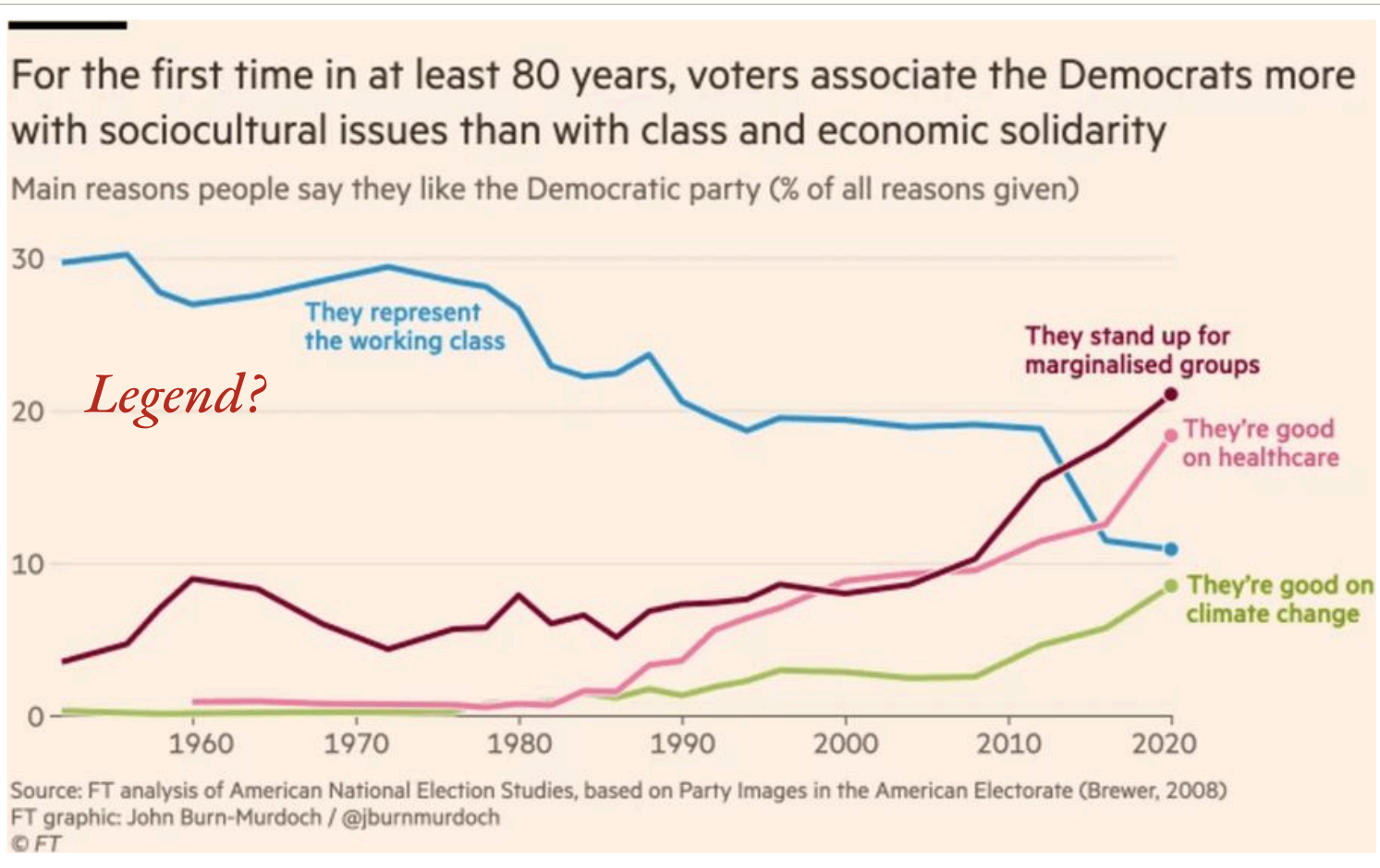
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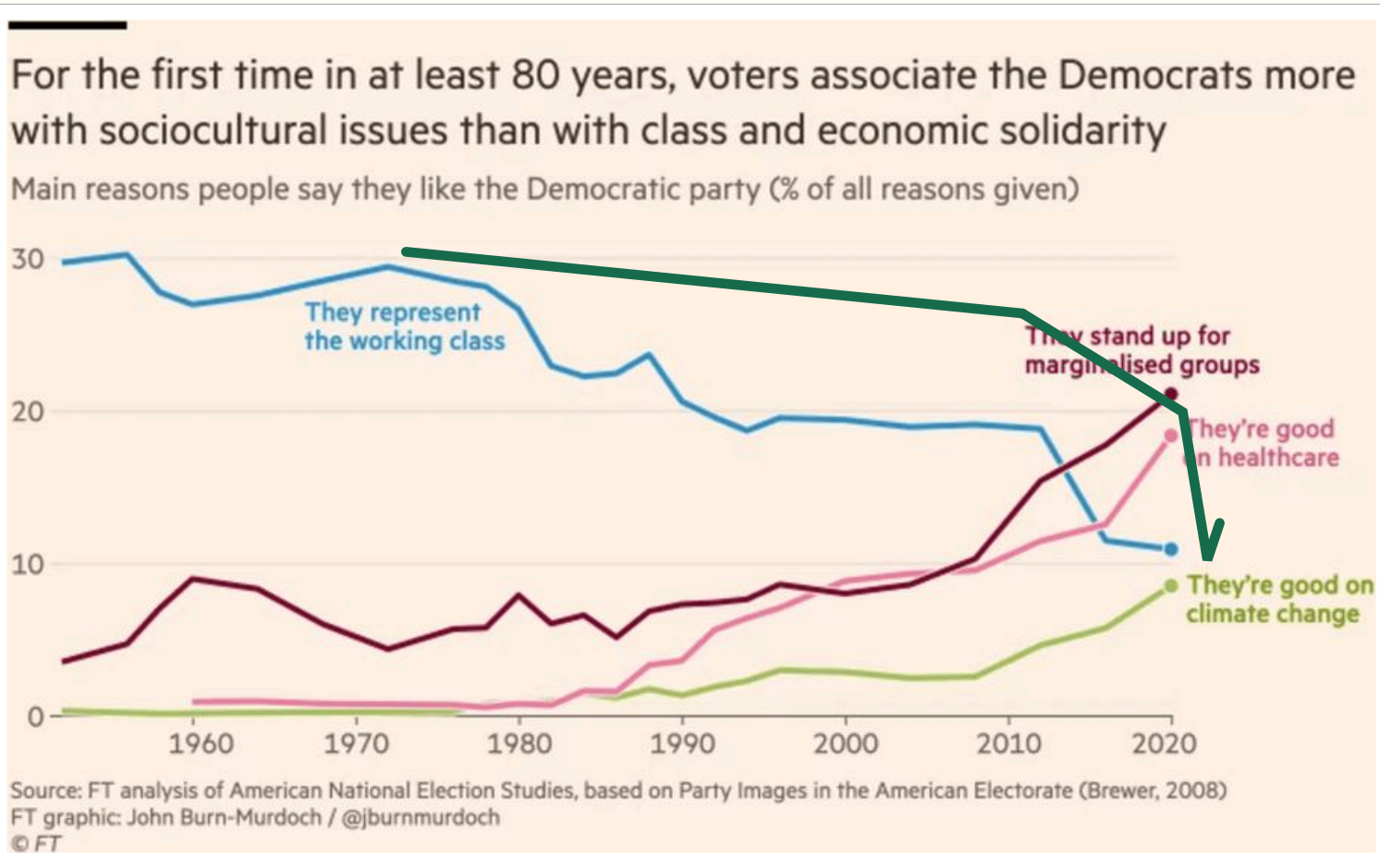


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FT graphic: John Burn-Murdoch / @burnmurdoch
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Start simple:

- *No color*
- *Get layout and
content right*

Start simple:

- *No color*
- *Get layout and content right*

Then add detail:

- *Color*
- *Animation*
- *Callouts*
- *etc...*

*MAKE THE
FONT BIGGER*

*MAKE THE
FONT BIGGER*

And not too silly

*MAKE THE
FONT BIGGER*

And not too silly or mixed

Regarding Fonts

*Sans-serif may be easier to read on screen
modern, clean, minimalist, “techy”*

Serif better in print: traditional, authoritative

Make the big idea clear at a glance.

Make the big idea clear at a glance.
Detail on inspection

Make the big idea clear at a glance.

Detail on inspection

(remember context and audience)